

A Simple and Unified Approach for Proving Knowledge of Isogenies between Abelian Varieties

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Abstract. In this paper we introduce a simple and unified approach, using generic proof systems, to prove knowledge of any isogeny between principally polarized abelian varieties of any dimension, assuming the 2^m -torsion is accessible for sufficiently large m . Prior generic approaches were restricted to smooth degree isogenies between elliptic curves, requiring bespoke constraints for each small prime factor ℓ of the degree, typically derived from (a variant of) the ℓ -th modular polynomial.

Our approach instead relies only on proving knowledge of a 2^n -isogeny between principally polarized abelian varieties, with constraints that are essentially the same in every dimension, yielding a uniform and easier-to-optimize algorithm. Since any isogeny can be embedded in a 2^n -isogeny by increasing the dimension (a result of Robert), and conversely a 2^n -isogeny between products of varieties yields an isogeny of degree $\leq 2^n$ between a factor of the domain and codomain, our construction immediately proves knowledge of an isogeny of any degree between two elliptic curves via a higher-dimensional representation; it does so without disclosing the degree or requiring knowledge of the endomorphism ring, thereby solving an open problem posed by Beullens, De Feo, Galbraith, and Petit in 2023.

Two use cases are immediate: proving knowledge of any isogeny between two supersingular curves over $\mathbb{F}_{p^2}$ (e.g. generating a curve with unknown endomorphism ring), and proving knowledge of a class-group-action isogeny on oriented supersingular curves over $\mathbb{F}_p$ (e.g. CSIDH), which **qt-Pegasis** naturally represents in dimension 4.

Finally, we propose two tailored zero-knowledge proof systems improving proving time and proof size without loss of generality, and provide the first implementation in dimensions 2 and 4, in Rust.

Keywords: Post-quantum cryptography · isogenies · zero-knowledge proofs.

1 Introduction

The task of proving knowledge of a secret isogeny goes back to the very beginning of isogeny based cryptography. Already in the seminal paper by Couveignes [25], an ID-protocol proved knowledge of a secret isogeny coming from the class group action on ordinary elliptic curves; moving to oriented supersingular curves

later gave the much more efficient CSIDH [18], to which the same ID-protocol applies as a simple variant of the graph isomorphism proof of Goldreich, Micali, and Wigderson [43]. It is a sigma protocol with binary challenge space, requiring λ repetitions for λ -bit security, and requires acting by random class group elements. Until recently, this was only possible via workarounds such as rejection sampling (SeaSign [33]) or computing the class group structure (CSI-FiSh [12]). Since Clapote [58] and its more efficient variants Pegasis [29] and qt-Pegasis [30], random class group elements can be acted with efficiently and the ID-protocol applies in its original form.

Proving knowledge of a (non-oriented) isogeny of *secret* degree between supersingular elliptic curves has proven much harder; the main example is the ID-protocol underlying (1-dimensional) SQIsign [35, 36, 24], which however requires significant setup and knowledge of both endomorphism rings of domain and codomain.

On the other hand, proving knowledge of an isogeny of *smooth and known degree* between supersingular elliptic curves has many more constructions, both isogeny-tailored and generic. The first such ID-scheme was the original SIDH-paper by De Feo, Jao and Plût [34] (DFJP); a soundness error found by Ghan-tous, Pintore, and Veroni [42] was later fixed by De Feo, Dobson, Galbraith, and Zobernig [32], and despite being “based on SIDH” the protocol does not reveal torsion information and remains believed secure after the SIDH attacks [15, 54, 60]. It was generalized to any characteristic with statistical zero-knowledge in [6], which, like the class-group ID-protocol, suffers from a small challenge space requiring parallel executions.

A recent line of work instead uses generic zero-knowledge techniques such as zk-SNARKs to prove the existence of an isogeny between elliptic curves. Inspired by [21], Cong, Lai, and Levin [22] expressed a 2^n -isogeny as a sequence of j -invariant tuples satisfying the 2nd modular polynomial Φ_2 , turned this into R1CS constraints (with conditions avoiding backtracking), and applied generic zk-SNARKs (Aurora, Ligerio, Limbo). This approach is limited to degree 2^n isogenies of known degree; using modular polynomials Φ_ℓ for $\ell > 2$, including more compact canonical [44], Atkin, or Weber variants [45], extends this to smooth degrees with primes up to $\ell = 71$, but each new prime factor requires tailored constraints that scale poorly with ℓ even after weighting by $\log_2(\ell)$. Levin and Pedersen [49] replaced the modular polynomial with radical isogenies, avoiding backtracking and simplifying the constraints.

After the SIDH-attacks, higher-dimensional isogeny representations have become ubiquitous: SQIsign’s efficiency was massively improved by embedding isogenies in 2^n -isogenies between abelian surfaces, making it a strong post-quantum standardization candidate [1, 7, 14] (also changing it from a secret-degree to a known-but-non-smooth-degree proof); similarly, unrestricted class group action computations, needed for Couveignes’ ID-protocol [25], were only recently made efficient by embedding in a 2^n -isogeny between 4-dimensional abelian varieties [29, 30]. It is thus natural to prove knowledge of secret 2^n -isogenies between higher dimensional abelian varieties directly, which implies a proof of knowledge

for an isogeny between elliptic curve factors of the domain and codomain. This has two immediate advantages: the resulting constraints are simple and uniform across dimensions, and the embedding hides the (factors of the) degree of the secret isogeny.

Our contributions We provide the first simple and unified framework for proving knowledge of a chain of 2-isogenies in the level-2 theta model between abelian varieties in any dimension, using generic proof techniques. Since any N -isogeny between principally polarized abelian varieties can be embedded in a 2^n -isogeny of higher dimension, for some $2^n > N$ (see Lemma 2), we effectively present the first unified way of proving knowledge of any isogeny, regardless of its degree. Our main contributions can be summarized as follows:

- We show that the radical isogeny framework in the theta model introduced in [48] can be efficiently translated into an R1CS constraint system (Section 3); furthermore, the type of constraints is the same in all dimensions, and is also the same for all types of isogenies (e.g. so-called gluing, splitting, diagonal or generic), resulting in a very simple and unified framework (Section 5);
- Motivated by the embedding lemma (see Lemma 2), we analyze the extension of the radical framework to dimension 4 (in particular, we discuss the additional relations in Appendix A.1), and discuss its security (Section 4);
- We describe a tailor-made multilinear polynomial interactive oracle proof which can be compiled into a zk-SNARK using the BaseFold [66] polynomial commitment scheme without losing any generality (Section 6.1);
- We propose a lattice-based sigma protocol that has significantly smaller proof size and lower prover computation time at the cost of adding an additional hardness assumption (Section 6.2). We note that this proof system forms an independent contribution as a zero-knowledge post-quantum secure proof system for proving small statements over any field with small proof size;
- We solve one of the major open problems in [11], namely whether it is possible to prove knowledge of an arbitrary isogeny between two elliptic curves, without knowledge of the endomorphism ring of the starting curve and without leaking the degree of the secret isogeny; moreover, we discuss how our framework can be adapted to prove all the relations suggested in [11], showing its generality and flexibility (Section 7);
- We show the impact of our work on the recently growing field of higher dimensional isogeny constructions; notably, we give an alternative way of proving a qt-Pegasis isogeny, and we repair the 2-dimensional version of the CGL hash function (among other things, one of the main motivations behind [48]) from the recent attack of [16], by presenting a 2-dimensional trusted setup (Section 7);
- We provide a full implementation of our framework: we show how to augment the most common isogeny computation libraries in dimension 1, 2 and 4 to produce an R1CS transcript, and present a Rust implementation of our construction for proving and verifying such transcript (Section 8). Our implementation is available at

Technical overview The goal of this paper is to develop a proof of knowledge for the very general relation

$$\mathcal{R}_{\text{isog}(g)} := \left\{ ((A_0, A_1), \phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ \phi: A_0 \rightarrow A_1 \text{ is an arbitrary } N\text{-isogeny for some } N \in \mathbb{N} \end{array} \right\}.$$

In most practical applications, we will be interested in the above relation for $g = 1$, i.e. the A_i are taken to be elliptic curves, usually simply denoted $\mathcal{R}_{\text{isog}}$ (e.g. [11]). To achieve this, we show it is sufficient to give a proof of knowledge for the relation

$$\mathcal{R}_{\text{isog}(2^n, g)} := \left\{ ((A_0, A_1), \phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ \phi: A_0 \rightarrow A_1 \text{ is a good } 2^n\text{-isogeny} \end{array} \right\},$$

where a good 2^n -isogeny means that its kernel is isomorphic to $(\mathbb{Z}/2^n\mathbb{Z})^g$. Counter-intuitively, but by now a standard fact in isogeny-based cryptography, $\mathcal{R}_{\text{isog}(2^n, g)}$ is in fact general enough to encompass all instances of $\mathcal{R}_{\text{isog}(g)}$, possibly by increasing 2^n and g : given an (A_0, A_1) in the language of isogenous principally polarized abelian varieties of dimension g , a witness ϕ for $\mathcal{R}_{\text{isog}(2^n, g)}$ is clearly also a witness for $\mathcal{R}_{\text{isog}(g)}$, however given a witness ψ for $\mathcal{R}_{\text{isog}(g)}$, this can also be turned into a witness Ψ for $(A_0 \times B, A_1 \times B')$ in $\mathcal{R}_{\text{isog}(2^n, 8g)}$, for $2^n > \deg \psi$, and some auxiliary varieties B, B' , by Robert's embedding lemma (see Lemma 2). The isogeny $\Psi: A_0 \times B \rightarrow A_1 \times B'$ now again allows the extraction of a witness for (A_0, A_1) in $\mathcal{R}_{\text{isog}(g)}$ by composing with the projection maps.

To derive a proof of knowledge for the relation $\mathcal{R}_{\text{isog}(2^n, g)}$, we first arithmetize it using the level-2 theta model: [48] proves that a good 2^n -isogeny can be computed as a composition of 2-isogenies in the theta model, where the theta null points all satisfy the simple relation

$$\mathcal{S} \circ \mathcal{H}(\theta^{A_{i+1}}(0_{A_{i+1}})) = \mathcal{H} \circ \mathcal{S}(\theta^{A_i}(0_{A_i})),$$

where $\mathcal{S}$ denotes the coordinate-wise squaring map, and $\mathcal{H}$ denotes a Hadamard-transform. This turns a witness for $\mathcal{R}_{\text{isog}(2^n, g)}$ into a witness for an analogous relation $\mathcal{R}_{\text{HS}(n, g)}$ over such chains of theta null points, which [48] shows to be equivalent to $\mathcal{R}_{\text{isog}(2^n, g)}$ for $g \leq 2$. However, importantly, this is not true for $g > 2$, where extra relations are needed to certify that a witness for $\mathcal{R}_{\text{HS}(n, g)}$ corresponds to an actual isogeny (known for $g = 3$ from [48]; we derive the analogous relations for $g = 4$ in Appendix A.1, and show they incur significant overhead).

We argue instead that these extra relations can be dropped without compromising security. Writing $\tilde{\mathcal{R}}_{\text{HS}(n, g)} \subseteq \mathcal{R}_{\text{HS}(n, g)}$ for the subset of witnesses that do correspond to isogenies, we give heuristic evidence (Section 4) that finding a witness for $\mathcal{R}_{\text{HS}(n, g)}$ that is not a witness for $\tilde{\mathcal{R}}_{\text{HS}(n, g)}$ is computationally hard. Under this assumption, $\mathcal{R}_{\text{isog}(2^n, g)}$ and $\mathcal{R}_{\text{HS}(n, g)}$ remain weakly equivalent for any g , so it suffices to give a proof of knowledge for $\mathcal{R}_{\text{HS}(n, g)}$.

Remark 1. Although any NP statement can be proven in zero-knowledge via a generic program representation [43], doing so is far less efficient and, since a new arithmetization would be needed for every dimension (e.g. computing isogenies in dimension 8), far less uniform. Our hardness assumption on $\mathcal{R}_{\text{HS}(n,g)} \setminus \tilde{\mathcal{R}}_{\text{HS}(n,g)}$ therefore trades a mild extra assumption for both efficiency and uniformity across dimensions; we believe the assumption to be true, and disproving it is an interesting open problem.

Finally, we show how to prove knowledge of a witness for the relation $\mathcal{R}_{\text{HS}(n,g)}$. We do this by translating the Hadamard relations into a constraint system $M_1 z \circ M_1 z = M_2 z$ for public matrices M_1, M_2 and witness z . To prove this constraint system we define multilinear polynomials $\tilde{z}_1, \tilde{z}_2$ such that the previous relation holds if and only if $\forall X \in \{0, 1\}^m : \tilde{z}_1(X)^2 = \tilde{z}_2(X)$. Then we use two subsequent instantiations of the sumcheck protocol [50] to reduce this statement to proving the evaluation of a multilinear polynomial, for which we use the Base-Fold [66] polynomial commitment scheme. We further propose a trade-off where we decrease the proving time and proof size at the cost of adding well-known post-quantum lattice-based hardness assumptions. The resulting protocol recursively proves knowledge of valid sumcheck transcripts by committing to them in BDLOP commitments [8] and then using a lattice-based sigma protocol to prove linear relations. Non-linear verification steps in the sumcheck protocol can be masked using Laurent polynomials and then verified in the clear. Verifying the masking itself can again be expressed using linear relations.

Our resulting proof systems are general enough to derive proofs of knowledge for many relations of interest in isogeny-based cryptography. One example of this is the CSIDH-relation

$$\mathcal{R}_{\text{CSIDH}} := \{((E_0, E_1), \phi) \mid \phi : E_0 \rightarrow E_1 \text{ is an isogeny defined over } \mathbb{F}_p\}.$$

Compared to the state of the art, running the original ID-protocol by Couveignes, using qt-Pegasis [30], our generic proofs achieve 5.2s prover time with 1.1MiB proof sizes. This shows that our proof system, although very generic, is already quite practical for many applications, with a performance that is comparable to specialized native protocols.

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2 Preliminaries

In this work, we consider isogenies between principally polarized abelian varieties equipped with a level-2 theta structure, and zero-knowledge proofs of knowledge of such maps. We recall the necessary background on such varieties, their relation to computing (good) 2^n -isogenies, and on zero-knowledge proof systems; for a more general introduction to (principally polarized) abelian varieties, see Milne [55].

2.1 Abelian varieties and isogenies

An abelian variety is a smooth, projective algebraic variety equipped with a commutative group law. Abelian varieties of dimension 1 are elliptic curves. The g -th power of an elliptic curve E^g is an example of a g -dimensional abelian variety. When not clear from the context, we use $A(k)$ to denote the set of k -rational points of A . $A[\ell](k)$ denotes the ℓ -torsion of that set, i.e. the k -rational points of order ℓ . Throughout the rest of this work, p will be a prime, and we will be mostly interested in $k = \mathbb{F}_{p^2}$ or $\mathbb{F}_p$, thus dropping the k from above.

In particular, in dimension 1 we will focus on *supersingular* elliptic curves, i.e. curves such that $E[p] = \{0_E\}$. One important property of such curves is that they always admit a model which is defined over $\mathbb{F}_{p^2}$. One generalization of supersingularity is that of *superspecial* abelian varieties. Superspecial abelian varieties admit a model over $\mathbb{F}_{p^2}$ as well [17]. Products of elliptic curves and all other abelian varieties in this work will be superspecial.

An isogeny is a surjective morphism between abelian varieties, respecting the group law, with finite kernel. *Separable* isogenies can be identified by their kernel, up to post-composition with an isomorphism. For every abelian variety A , there is a corresponding dual variety $\hat{A}$. An isogeny $\lambda : A \rightarrow \hat{A}$ is called a *polarization* if it is induced by an ample line bundle [55, Section I.11] (over $\bar{k}$ in general, though this distinction is not necessary when k is a finite field [23, Theorem 2.6]); if it is an isomorphism the polarization is *principal*. A pair (A, λ) where λ is a principal polarization of A is called a *principally polarized abelian variety* (PPAV). A principal polarization induces a polarized Weil-pairing $e_N^\lambda : A[N] \times A[N] \rightarrow \mu_N$, with μ_N the N -th roots of unity, by precomposing the standard Weil-pairing with the polarization.

An isogeny $\phi : (A, \lambda_A) \rightarrow (B, \lambda_B)$ is called compatible with the polarization if $\hat{\phi} \circ \lambda_B \circ \phi = [N]\lambda_A$ for some N . Such a compatible isogeny is called an N -isogeny. For especially nice N -isogenies, we use the following definition:

Definition 1. *An N -isogeny $\phi : A \rightarrow B$ between principally polarized abelian varieties of dimension g is called a good N -isogeny if $\ker \phi \simeq (\mathbb{Z}/N\mathbb{Z})^g$.*

A good N -isogeny is more commonly denoted as an (N, N) -isogeny in dimension 2, an (N, N, N) -isogeny in dimension 3 and so on.

2.2 Good extensions

Let $\phi : A \rightarrow B$ and $\psi : B \rightarrow C$ be two good N -isogenies, for some prime N , between PPAVs of dimension g . If $g = 1$, the composition $\psi \circ \phi$ can be either a good N^2 -isogeny (with kernel $\cong \mathbb{Z}/N^2\mathbb{Z}$) or multiplication by N (with kernel $\cong (\mathbb{Z}/N\mathbb{Z})^2$). In the latter case $\psi = \hat{\phi}$ is the *dual* isogeny of ϕ (recall that in dimension 1, the polarization is canonical). The situation becomes more complex for $g \geq 2$: in general we have $\ker(\psi \circ \phi) \cong (\mathbb{Z}/N^2\mathbb{Z})^m \times (\mathbb{Z}/N\mathbb{Z})^{2k}$ with $m + k = g$. This motivates the following definition, originally due to [17] and later generalized by [48] to arbitrary g .

Definition 2. Let $\phi : A \rightarrow B$ and $\psi : B \rightarrow C$ be good N -isogenies between principally polarized abelian varieties of dimension g , and $\ker(\psi \circ \phi)$ as above. Then:

- if $k = g$ and $m = 0$, we say that ψ is the *dual extension* of ϕ ;
- if $k = 0$ and $m = g$, we say that ψ is a *good extension* of ϕ ;
- otherwise, we say that ψ is a *bad extension* of ϕ .

Good extensions can also be characterized as isogenies ψ such that $(\ker \psi) \cap \phi(A[N]) = 0$ [17]. For a given good N -isogeny ϕ , there are $N^{g(g+1)/2}$ good extensions [19, Lemma 2]. Composing ϕ with any good extension results in a good N^2 -isogeny. In this work, we will be working with radical isogenies. Isogenies built in this way are automatically good extensions [48].

The dual extension of ϕ is the isogeny such that $\ker \psi = \phi(A[N])$, and corresponds to the (polarized) dual of ϕ .

2.3 Embedding isogenies

One of the main outcomes of the SIDH attacks [15, 54, 60] is that one-dimensional isogenies can be *embedded* in higher dimensional isogenies. The main building block that powers this technique is the now famous Kani’s Lemma [46]:

Lemma 1. Given a commutative diagram of isogenies between principally polarized abelian varieties

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow \psi & & \downarrow \psi' \\ A' & \xrightarrow{\phi'} & B' \end{array}$$

where ϕ, ϕ' are both n_1 -isogenies, and ψ, ψ' are n_2 -isogenies, the map

$$\Phi = \begin{pmatrix} \phi & \tilde{\psi}' \\ -\psi & \phi' \end{pmatrix} : A \times B' \rightarrow B \times A'$$

is an N -isogeny, for $N = n_1 + n_2$.

Proof. See [59, Lemma 2.3] for this formulation. □

Clearly, being able to evaluate Φ is sufficient for being able to evaluate ϕ (and ψ, ϕ' and ψ'), by composing with the appropriate injection/projection maps. Hence, we say that these isogenies are *embedded* in Φ .

The reason why we are interested in this constructively is that the complexity of evaluating an isogeny when given its kernel depends (exponentially) on the largest prime factor of its degree. If n_1 contains a large prime factor but $N = n_1 + n_2$ is smooth, Φ and hence ϕ can be evaluated efficiently at the cost of doubling the dimension.

For efficiency reasons one is generally interested in evaluating a one-dimensional isogeny ϕ of degree n_1 , and one fixes $N = 2^n$. Without any knowledge of the endomorphism ring of the starting curve A , building an isogeny ψ of the correct degree $2^n - n_1$ (sometimes called the *auxiliary isogeny*) is a challenging task on its own. However, we always know $\mathbb{Z} \subset \text{End}(E)$, i.e. we can always use scalar multiplication as an endomorphism. We have the following cases: 1) if n_1 is smooth, ϕ can already be evaluated in dimension 1; 2) if $2^n - n_1 = a^2$ is a square, then taking $\psi = [a]$ the multiplication by a map is a valid choice; 3) if $2^n - n_1 = a^2 + b^2$ is a sum of two squares, then

$$\psi = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is an endomorphism of $E \times E$ of the correct degree; we must hence work in dimension 4; 4) due to a famous result by Lagrange, every integer can be written as a sum of four squares; thus we can always write an endomorphism of E^4 consisting of integer entries and use it to embed any isogeny in dimension 8.

This leads to the following lemma, due to Robert [59, Section 3].

Lemma 2. *Let $\phi : A \rightarrow B$ be an N -isogeny between principally polarized abelian varieties. Then ϕ can be embedded in an N' -isogeny*

$$\Phi : A^4 \times B^4 \rightarrow A^4 \times B^4$$

for any $N' > N$.

2.4 Level-2 theta structures

Given a principally polarized abelian variety A of dimension g , a *level-2 symmetric theta structure* on A is in particular a map $\theta^A : A \rightarrow \mathbb{P}^{2^g-1}$, which induces a symplectic basis (with respect to the polarized Weil-pairing) of $A[2]$. Such a symplectic basis of $A[2] \simeq (\mathbb{Z}/2\mathbb{Z})^{2g}$ is given by $2g$ 2-torsion points $\{P_1, \dots, P_g, Q_1, \dots, Q_g\}$ such that $e_2^\lambda(P_i, P_j) = e_2^\lambda(Q_i, Q_j) = 1$ and $e_2^\lambda(P_i, Q_j) = (-1)^{\delta_{i,j}}$ with $\delta_{i,j}$ Kronecker delta for all $1 \leq i, j \leq g$.

We now give the following full definition, due to Duparc [38, Section 2.2], which the reader may find simpler than the original definition by Mumford.

First, note that a symplectic basis of $A[2]$ induces an isomorphism

$$\pi : A[2] \simeq (\mathbb{Z}/2\mathbb{Z})^g \times \widehat{(\mathbb{Z}/2\mathbb{Z})^g}$$

(called a symplectic representation) such that $e_2^\lambda(P, Q) = (-1)^{\widehat{x_Q}(x_P) - \widehat{x_P}(x_Q)}$ for all $P, Q \in A[2]$ (here, we denote $\pi(R) = (x_R, \widehat{x_R})$). This allows the following definition.

Definition 3. *Let A be a principally polarized abelian variety of dimension g . A (symmetric) level-2 theta structure is a map $\theta^A : A \rightarrow \mathbb{P}^{2^g-1}$, whose coordinates we denote by θ_i^A indexing by $i \in (\mathbb{Z}/2\mathbb{Z})^g$, which satisfies*

$$\theta_i^A(P + Q) = (-1)^{\widehat{x_Q}(i)} \theta_{i+x_Q}^A(P) \quad (1)$$

for all $P \in A$ and $Q \in A[2]$.

Informally, a level-2 theta structure thus corresponds to a map θ from A to the projective space, such that the 2-torsion acts by very specific linear transformations on the points of A . Note that this map is not injective. However, when A is irreducible, it does induce an injection of the Kummer variety $A/[\pm 1] \hookrightarrow \mathbb{P}^{2^g-1}$.

An important invariant of a principally polarized abelian variety with a level-2 theta structure is the *theta null point* $\theta^A(0_A)$. However, note that this is not an isomorphism invariant of A as a principally polarized abelian variety alone (i.e. seen without a theta structure): different choices of theta-structure give different theta null points for the same A . Note also that a level-2 theta structure induces the symplectic basis of $A[2] = K_1 + K_2$ (which is referred to as *the canonical decomposition*); this basis is easily recovered from the theta null point by applying Equation (1) with $P = 0_A$ (for an explicit description, see [48, Lemma 1]).

Formulae for the computation of 2^n -isogenies in the level-2 theta model in dimension 2 were originally given in [31] (while dimension 4 was done by Dartois [27]). As the theta-structure already induces two choices of maximally isotropic subgroup K_1, K_2 (from $A[2] = K_1 + K_2$), a 2-isogeny $\phi : (A, \theta^A) \rightarrow (B, \theta^B)$ will refer to an isogeny with kernel K_2 , such that θ^B gives a new kernel, defining a good extension of ϕ . To describe the action of these isogenies on the theta null points, we recall the notation

$$\mathcal{S} : \mathbb{P}^{2^g-1} \rightarrow \mathbb{P}^{2^g-1} : (x_0 : \dots : x_{2^g-1}) \rightarrow (x_0^2 : \dots : x_{2^g-1}^2)$$

denoting coordinate-wise squaring, and

$$\mathcal{H} : \mathbb{P}^{2^g-1} \rightarrow \mathbb{P}^{2^g-1} : (x_0 : \dots : x_{2^g-1}) \rightarrow H_g(x_0, \dots, x_{2^g-1})^T$$

where

$$H_g := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{\otimes g}$$

is the g -th Hadamard transform. This gives the following lemma [48, Lemma 3]:

Lemma 3. *Let (A, θ^A) be an abelian variety with a level-2 theta structure, and let $A[2] = K_1 + K_2$ be the canonical decomposition. Let $\phi : (A, \theta^A) \rightarrow (B, \theta^B)$*

be a 2-isogeny, with kernel K_2 , where θ^B is a compatible theta structure on B . Then

$$\mathcal{S} \circ \mathcal{H}(\theta^B(0_B)) = \mathcal{H} \circ \mathcal{S}(\theta^A(0_A)). \quad (2)$$

Further, there are $g(g+1)/2$ free choices of square-roots corresponding to the different compatible theta-structures on B .

In [48, Lemma 3], the authors give more details on the choices of square-roots, but the description above will be sufficient for us. One important detail is that composing 2-isogenies obtained by Lemma 3 automatically gives good extensions: see [48, Section 3.4].

Since $\mathcal{H}$ is invertible, it is clear that once we fix θ^A , there are $2^g - 1$ different choices of θ^B satisfying the equation (corresponding to the different choices of square roots, projectively). In particular, for dimension $g = 1, 2$, we have $g(g+1)/2 = 2^g - 1$, and thus the number of possible square root choices and compatible theta-structures are the same. Hence, any choice is valid and corresponds to the different possible good extensions.¹

In fact, the relation in Lemma 3 comes from a more general formula for the evaluation of any theta-point [48, Section 2.5]: given an isogeny ϕ as in Lemma 3, and a point $P \in A$, we have the relation

$$\mathcal{H}(\theta^B(\phi(P))) \odot \mathcal{H}(\theta^B(0_B)) = \mathcal{H} \circ \mathcal{S}(\theta^A(P)), \quad (3)$$

where $\odot$ denotes coordinate-wise multiplication. Notice that this relation specializes to Equation (2) for $P = 0_A$.

2.5 Proof systems

Let $\mathcal{R} : X \times W \rightarrow \{0, 1\}$ be a relation between the input set X and the witness set W , defining the NP-language $\mathcal{L} = \{x \in X \mid \exists w \in W \text{ s. t. } (x, w) \in \mathcal{R}\}$.

Lemma 4 (Schwartz-Zippel Lemma). *Let $f \in \mathbb{F}[X_1, \dots, X_m]$ be a non-zero polynomial of total degree d over a field $\mathbb{F}$. Let S be any finite subset of $\mathbb{F}$, and let $r_1, \dots, r_m$ be m field elements selected independently and uniformly from the set S , then*

$$\Pr[f(r_1, \dots, r_m) = 0] \leq \frac{d}{|S|}.$$

Definition 4 (Multilinear extensions). *For every function $f : \{0, 1\}^m \rightarrow \mathbb{F}$, there is a unique multilinear polynomial $\tilde{f} \in \mathbb{F}[X_1, \dots, X_m]$ such that $\forall b \in \{0, 1\}^m : \tilde{f}(b) = f(b)$. We call $\tilde{f}$ the multilinear extension of f and $\tilde{f}$ can be expressed as*

$$\tilde{f}(X) = \sum_{b \in \{0, 1\}^m} f(b) \cdot \tilde{\mathbf{eq}}(b, X)$$

¹More precisely, any choice of square root yields the same isogeny up to isomorphism, but different choices determine different good extensions in the following two steps.

where $\text{eq} : \{0, 1\}^m \times \{0, 1\}^m \rightarrow \{0, 1\} : (x, y) \mapsto \mathbb{1}[x = y]$. We can compute

$$\tilde{\text{eq}}(x, y) = \prod_{i=1}^m (x_i y_i + (1 - x_i)(1 - y_i)).$$

Definition 5 (Sumcheck protocol [50]). For a multivariate polynomial $f \in \mathbb{F}[X_1, \dots, X_m]$ with m variables with degree at most ℓ in each variable and some $S \in \mathbb{F}$, the sumcheck protocol is a public-coin interactive protocol between a prover P and a verifier V that reduces the claim $S \stackrel{?}{=} \sum_{b \in \{0, 1\}^m} f(b)$ to the claim that $f(r) \stackrel{?}{=} e$ for some uniformly random $e \in \mathbb{F}$ and $r \in \mathbb{F}^m$. The protocol has $\ell \cdot m / |\mathbb{F}|$ soundness error and $\mathcal{O}(\ell \cdot m)$ communication cost.

3 Path of theta null points

In this section, we discuss the relationship between knowing a path of level-2 theta null points satisfying Equation (2), and knowing a 2^n -isogeny between abelian varieties of dimension g . We also show how Equation (3) can be used to additionally derive relations for the evaluation of points. Most of the material in this section follows from [48], but will be essential to our generic proofs. Of particular importance is Definition 6, and Problems 1 and 2, which formalize the issue that a path of theta null points is only equivalent to an isogeny-path when $g \leq 2$.

3.1 Encoding an isogeny-path as a set of solutions

Consider a good 2^n -isogeny expressed as a composition of n good 2-isogenies

$$A_0 \xrightarrow{\phi_0} A_1 \xrightarrow{\phi_1} \dots \xrightarrow{\phi_{n-1}} A_n$$

between abelian varieties A_i of dimension g . We note that in practical applications, e.g. when the 2^n -isogeny corresponds to a 4-dimensional representation of a class group action, both A_0 and A_n typically are products of abelian varieties of lower dimension. Since the connecting isogenies however are still good, these gluing (at the start) and splitting isogenies (at the end) are part of the above diagram and do not require separate treatment. This is in stark contrast to the actual computation of these isogenies, which requires tailored functions for each type of isogeny.

By Lemma 3, there exist level-2 theta structures θ^{A_i} on A_i for all i , such that the theta null points all satisfy

$$\mathcal{S} \circ \mathcal{H}(\theta^{A_{i+1}}(0_{A_{i+1}})) = \mathcal{H} \circ \mathcal{S}(\theta^{A_i}(0_{A_i})). \quad (4)$$

The whole path can therefore be expressed as a concatenation of $(n + 1)$ theta null points, each consisting of 2^g finite field elements. More in detail, the path is given by the set of $2^g \times (n + 1)$ values X_i^j for $0 \leq i \leq 2^g - 1$, $0 \leq j \leq n$ such that

$X^j = (X_0^j : \dots : X_{2^g-1}^j) = \theta^{A_j}(0_{A_j})$. The first and last varieties of the chain A_0 and A_n , and consequently X^0, X^n , are public, so the variables X_i^j satisfy the following relations

$$\begin{cases} X^0 = \theta^{A_0}(0_{A_0}), \\ \mathcal{S} \circ \mathcal{H}(X^{j+1}) = \mathcal{H} \circ \mathcal{S}(X^j), \quad 0 \leq j \leq n-1, \\ X^n = \theta^{A_n}(0_{A_n}). \end{cases} \quad (5)$$

Note that since theta null points are projective, the above equalities also have to be interpreted as such, i.e. they allow scaling each theta null point by a different non-zero scalar. By normalizing the first theta null point $\theta^{A_0}(0_{A_0})$ and imposing that the above equalities hold exactly (and not just projectively), we will impose a specific representative for each following theta null point $\theta^{A_i}(0_{A_i})$.

Given a good 2^n -isogeny, it is almost trivial to recover the corresponding theta null points $\theta^{A_i}(0_{A_i})$ and thus the full path X_i^j : they are induced by any symplectic basis $A_i[2] = K_1 + K_2$, where K_2 corresponds to the kernel of the isogeny ϕ_i . More details are given in Section 8.1.

Remark 2. There is a subtlety in going from an efficient representation of a 2^n -isogeny $\phi : A_0 \rightarrow A_n$, to a chain of 2-isogenies $\phi_{n-1} \circ \dots \circ \phi_0 = \phi$, as this is not an easy task in general. In order to factor the isogeny, we need access to the kernel $K \subset A_0[2^n]$. This can be obtained from the representation of ϕ , provided the 2^n -torsion is accessible, or even $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion, by working with the duals, and doing a meet-in-the-middle approach. This is why we will need to require that the 2^m -torsion is accessible for sufficiently large m in general.

3.2 Evaluation of points

We now extend the relation from the previous section to additionally prove the evaluation of a point through the isogeny.

Consider the chain from the previous section. Given a point $P_0 \in A_0$, set $Y^0 = \theta^{A_0}(P_0)$, write $Y^{j+1} = (Y_0^{j+1} : \dots : Y_{2^g-1}^{j+1}) = \theta^{A_{j+1}}(\phi_j(Y^j))$ for $0 \leq j \leq n-1$, and recall that $X^j = \theta^{A_j}(0_{A_j})$. By Equation (3), the coordinates satisfy

$$\mathcal{H}(Y^{j+1}) \odot \mathcal{H}(X^{j+1}) = \mathcal{H} \circ \mathcal{S}(Y^j), \quad (6)$$

where we recall that $\odot$ denotes coordinate-wise multiplication.

Thus, for a point $P_0 \in A_0$, and $P_n = \phi(P_0) \in A_n$, the point evaluation can be expressed by another $2^g \times (n+1)$ values Y_i^j , satisfying

$$\begin{cases} Y^0 = \theta^{A_0}(P_0), \\ \mathcal{H}(Y^{j+1}) \odot \mathcal{H}(X^{j+1}) = \mathcal{H} \circ \mathcal{S}(Y^j), \quad 0 \leq j \leq n-1, \\ Y^n = \theta^{A_n}(P_n) \end{cases} \quad (7)$$

It is crucial that the values X^{j+1} are the same as those in Equation (5). Their dual coordinates $\mathcal{H}(X^{j+1})$ determine the choice of square roots, and therefore ensure that Y^j is evaluated through the same isogeny as the one encoded by the null-point path.

Remark 3. We note that the last relation in Equation (7) may be replaced by other relations, depending on the purpose. If we are only given the evaluation up to scaling, then that secret scalar multiplication in the end must also give new relations. If we are only interested in proving that the evaluation is non-zero (see e.g. the $g = 1$ case in Section 4.2), the last relation may simply be replaced by $Y^n \neq \theta^{A_n}(0_{A_n})$.

3.3 Recovering an isogeny-path from a solution set

We now discuss how to recover a 2^n -isogeny from a set of solutions to Equation (5). As already mentioned, we are forced to split our discussion into two cases. Indeed, from Lemma 3, we see that if we fix A_i there are in total $2^g - 1$ solutions to Equation (4), of which only $g(g+1)/2$ correspond to valid theta structures of abelian varieties A_{i+1} . For $g = 1, 2$ these two quantities coincide, but for $g > 2$ this is no longer the case. We therefore make the following case distinction.

Dimension $g \leq 2$. The simplest case is dimension $g \leq 2$. Assume we are given the theta null points $\theta^{A_0}(0_{A_0}), \theta^{A_n}(0_{A_n})$, and we wish to recover the 2^n -isogeny between them, encoded by the set of solutions $T_1, \dots, T_{n-1} \in \mathbb{P}^{2^g-1}$, all satisfying Equation (5), where the system of equations holds affinely. As the solutions to

$$\mathcal{S} \circ \mathcal{H}(x_0 : \dots : x_{2^g-1}) = \mathcal{H} \circ \mathcal{S}(\theta^{A_0}(0_{A_0})) \quad (8)$$

correspond exactly to the different good extensions from (A_0, θ^{A_0}) (see Section 2.4), we know that any valid solution-set must correspond to a theta null point of a 2-isogenous variety, inducing a good extension. By induction, we can thus extract a good 2^n -isogeny from the solution set.

Dimension $g > 2$. The same argument fails as soon as $g > 2$. As mentioned before, there are $2^g - 1$ solutions to Equation (8) but only $g(g+1)/2$ correspond to valid isogenies. Thus, there are solutions to the above equations, which do not correspond to paths of 2-isogenies. To make this distinction clear, we introduce the following terminology.

Definition 6. Let $(A_0, \theta^{A_0}), (A_n, \theta^{A_n})$ be two isogenous abelian varieties of dimension g with a given level-2 theta-structure. Let $T_1, \dots, T_{n-1} \in \mathbb{P}^{2^g-1}$ be points satisfying Equation (5). We call the tuple $(\theta^{A_0}(0_{A_0}), T_1, \dots, T_{n-1}, \theta^{A_n}(0_{A_n}))$ a pseudo 2^n -isogeny path.

Computing a pseudo-isogeny path is formalized in the following problem.

Problem 1 (Pseudo-isogeny path problem). Given two isogenous abelian varieties of dimension g with a given level-2 theta structure (A_0, θ^{A_0}) and (A_n, θ^{A_n}) , find a pseudo 2^n -isogeny path between them for some n .

The previous problem can be seen as a natural generalization of the isogeny-problem in dimension $g > 2$. However, it seems unlikely that the above problem is significantly easier than the isogeny-problem itself, especially considering they are equivalent in dimension $g \leq 2$. Since we already assume the hardness of the isogeny-path problem, we are mainly interested in the hardness of computing a pseudo-isogeny path that does not correspond to an actual isogeny path.

Problem 2 (Non-isogeny path problem). Find points $T_0, T_1, \dots, T_n \in \mathbb{P}^{2g-1}$, such that $T_0 = \theta^{A_0}(0_{A_0})$ and $T_n = \theta^{A_n}(0_{A_n})$, where (A_0, θ^{A_0}) and (A_n, θ^{A_n}) are isogenous abelian varieties of dimension g with a given level-2 theta structure, but at least one pair T_i, T_{i+1} does not occur as the theta null points of 2-isogenous abelian varieties of dimension g .

We analyze the hardness of Problem 1 and Problem 2 in Section 4, though note already now that for $g \leq 2$, Problem 2 is not only a hard problem, but in fact an impossible problem.

In Appendix A we show that, at least in theory, additional constraints can be derived to assure for dimension $g > 2$ that a path really corresponds to a 2^n -isogeny. However, especially for $g \geq 4$, these constraints quickly become unwieldy and it would simply not be practical to explicitly include them. In the next section, we will argue that these extra constraints can in fact be left out without compromising security.

Remark 4. As mentioned in Remark 1, one could in principle prove an isogeny computation without assuming Problem 2, by proving in zero knowledge the circuit computing the isogeny; this would however cause a significant overhead. A more direct approach would be to prove knowledge of the kernel of the target isogeny. To do so, one needs to prove knowledge of a (secret) point P of the correct order on a (public) variety given in theta coordinates. Then at each step, the point of order 8 below P can be used to compute the next step and P can be pushed through the isogeny corresponding to such a step. Notice that if we are computing a 2^n -isogeny, at step m we need to compute $n - m - 2$ point doublings to derive a point of order 8 from P . On top of introducing various technicalities, this approach generates an amount of relations to prove which is quadratic in n (this can be reduced to $n \log n$ by using *optimal strategies*, see e.g. [1]), compared to the linear amount we obtain by assuming Problem 2.

4 Equivalence of isogeny relations

In the previous section, we discussed the following two relations

$$\mathcal{R}_{\text{isog}(2^n, g)} := \left\{ ((A_0, A_1), \phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ \phi: A_0 \rightarrow A_1 \text{ is a good } 2^n\text{-isogeny} \end{array} \right\}.$$

and

$$\mathcal{R}_{\text{HS}(n, g)} := \left\{ \left(\begin{array}{l} ((A_0, \theta_0), (A_n, \theta_n)), \\ (T_i)_{i \in [n-1]} \end{array} \right) \mid \begin{array}{l} (A_0, \theta_0), (A_n, \theta_n) \text{ isogenous abelian varieties of} \\ \text{dimension } g \text{ with a level-2 theta structure,} \\ S \circ \mathcal{H}(T_1) = \mathcal{H} \circ S(\theta^{A_0}(0_{A_0})), \\ S \circ \mathcal{H}(\theta^{A_n}(0_{A_n})) = \mathcal{H} \circ S(T_{n-1}), \\ S \circ \mathcal{H}(T_{i+1}) = \mathcal{H} \circ S(T_i) \quad \forall i \in [n-2] \end{array} \right\},$$

that are related to the difference of Problem 1 and Problem 2.

Additionally, we are interested in the very general relation

$$\mathcal{R}_{\text{isog}(g)} := \left\{ ((A_0, A_1), \phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ \phi: A_0 \rightarrow A_1 \text{ is an arbitrary } N\text{-isogeny for some } N \in \mathbb{N} \end{array} \right\}.$$

In Section 5 we will show how to give a zero-knowledge proof for $\mathcal{R}_{\text{HS}(n,g)}$. The goal for this section is to prove that (subrelations of) these relations are (weakly) equivalent, i.e. that there exists a polynomial time algorithm turning a witness for one into a witness for the other ones. For $\mathcal{R}_{\text{isog}(g)}$, we will see that this will additionally require a proof that the embedded isogeny has finite kernel.

Before doing so, we must be more precise about what we mean by witness. We assume to be working over a fixed finite field k (generally $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ for some prime p). Then a witness $(T_i)_{i \in [n-1]}$ to $\mathcal{R}_{\text{HS}(n,g)}$ is a set of $2^g \times (n-1)$ values in k satisfying Equation (5). A witness for $\mathcal{R}_{\text{isog}(g)}$ (resp. $\mathcal{R}_{\text{isog}(2^n, g)}$) is an isogeny (resp. of degree 2^n) defined over k , that can be described and evaluated on points of $A_0(k)$ in polynomial time. This goes by the name of *efficient isogeny representation* (see [61] for a survey).

4.1 Relation between $\mathcal{R}_{\text{isog}(g)}$ and $\mathcal{R}_{\text{isog}(2^n, g')}$

Our first goal will be to prove that $\mathcal{R}_{\text{isog}(2^n, g')}$ is general enough to express elements of $\mathcal{R}_{\text{isog}(g)}$. This is a direct consequence of Lemma 2.

We start with the following lemma.

Lemma 5. *There exists an algorithm, polynomial in $\log n$, that converts a witness ϕ for the statement (A_0, A_1) in $\mathcal{R}_{\text{isog}(g)}$ to a witness Φ for the statement $(A_0^4 \times A_1^4, A_0^4 \times A_1^4)$ in the relation $\mathcal{R}_{\text{isog}(2^n, 8g)}$ for any n such that $2^n > \deg(\phi)$.*

Proof. This is just a rephrasing of Lemma 2. $\square$

However, there is also a sense in which the corresponding relations in $\mathcal{R}_{\text{isog}(2^n, g)}$ give back relations in $\mathcal{R}_{\text{isog}(g)}$. To make this precise, we first define the subset $\mathcal{R}_{\text{isog}(g) \leq B} \subseteq \mathcal{R}_{\text{isog}(g)}$ as

$$\mathcal{R}_{\text{isog}(g) \leq B} := \{ ((A_0, A_1), \phi) \in \mathcal{R}_{\text{isog}(g)} \mid \phi \text{ is an } N\text{-isogeny, for } N \leq B \}.$$

We must also consider a subset of $\mathcal{R}_{\text{isog}(2^n, g)}$, which acts as witnesses for the same statements as in $\mathcal{R}_{\text{isog}(g)}$. To do this, we define the relation

$$\mathcal{S}_{\text{isog}(2^n, g, d)} := \left\{ ((A_0, A_1), \Phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ ((A_0 \times B_0, A_1 \times B_1), \Phi) \in \mathcal{R}_{\text{isog}(2^n, dg)} \text{ for some PPAVs} \\ B_0, B_1 \text{ and some isogeny } \Phi \end{array} \right\}$$

which injects into $\mathcal{R}_{\text{isog}(2^n, dg)}$ by fixing a choice of representatives (B_0, B_1) for each (A_0, A_1) .

However, it is **not** the case that any witness for (A_0, A_1) in $\mathcal{S}_{\text{isog}(2^n, g, d)}$ gives a witness for the same statement in the relation in $\mathcal{R}_{\text{isog}(g)}$: denoting the injection $\iota: A_0 \hookrightarrow A_0 \times B_0$ and projection $\pi: A_1 \times B_1 \rightarrow A_1$, the homomorphism $\pi \circ \Phi \circ \iota$

may have infinite kernel (i.e. it is not an isogeny). The easiest example of this is to take any two 2^n -isogenies $\phi : A_0 \rightarrow B_1$, and $\psi : A_1 \rightarrow B_0$, and consider the anti-diagonal

$$\begin{pmatrix} 0 & \widetilde{\psi} \\ \phi & 0 \end{pmatrix} : A_0 \times B_0 \rightarrow A_1 \times B_1.$$

Thus, consider

$$\mathcal{S}_{\text{isog}(2^n, g, d)}^+ := \left\{ ((A_0, A_1, \Pi), \Phi) \left| \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ ((A_0 \times B_0, A_1 \times B_1), \Phi) \in \mathcal{R}_{\text{isog}(2^n, dg)} \text{ for some PPAVs} \\ B_0, B_1 \text{ and some isogeny } \Phi, \\ \Pi \text{ is a zero-knowledge proof that } \pi \circ \Phi \circ \iota \text{ has finite kernel} \end{array} \right. \right\}.$$

Now, we have that any witness for (A_0, A_1, Π) would also be a witness for the relation (A_0, A_1) in $\mathcal{R}_{\text{isog}(g)}$ (and even $\mathcal{R}_{\text{isog}(g) \leq 2^n}$), since Φ encodes an efficient representation of an isogeny between A_0 and A_1 as one of its components.

4.2 Proof of finite kernel

The previous subsection shows that relations in $\mathcal{R}_{\text{isog}(g)}$ can be represented by a natural subset of $\mathcal{R}_{\text{isog}(2^n, g)}$, which we have denoted $\mathcal{S}_{\text{isog}(2^n, g, d)}^+$. To prove equivalence, we now discuss how to produce the proof Π of the finiteness of the relevant kernel.

Consider first the case $g = 1$. Given an element $((E_0, E_1), \Phi) \in \mathcal{S}_{\text{isog}(2^n, 1, d)}$, we have an upper bound 2^n on the degree of the embedded isogeny ϕ (assuming it is non-zero), and thus any $P \in E_0$ with order strictly greater than 2^n should satisfy $\phi(P) \neq 0_{E_1}$. Proving this can be done using the relations discussed in Section 3.2.

Consider now $g > 1$. In this case, the embedded map $\phi : A_0 \rightarrow A_1$ may have infinite kernel, while being non-zero. Thus, we need to prove that $\ker \phi$ is finite, or equivalently, that ϕ is surjective [55, Proposition 7.1], since A_0, A_1 are both g -dimensional. Again given a bound on $\deg \phi$, this can be done the following way: for any $M > \deg \phi$ fix public bases $\langle P_1, \dots, P_{2g} \rangle = A_0[M]$, and $\langle Q_1, \dots, Q_{2g} \rangle = A_1[M]$. The relations in Section 3.2, together with the formulae for e.g. addition can be used to prove knowledge of a matrix $T = (t_{ij})$ such that the action of ϕ on the M -torsion is given by this matrix relative to the chosen basis. Now, to check that ϕ is surjective, it suffices to prove that $\det(T) \not\equiv 0 \pmod{M}$, though notice that ϕ is not necessarily surjective on the M -torsion, so we may have $\det(T) \notin (\mathbb{Z}/M\mathbb{Z})^\times$. Notice also that we may compute T efficiently by simply setting $Q_1, \dots, Q_{2g}$ to be a random (known) change-of-basis of $\phi(P_1), \dots, \phi(P_{2g})$.

As a consequence we obtain the following lemma.

Lemma 6. *The relations $\mathcal{R}_{\text{isog}(g) \leq 2^n}$ and $\mathcal{S}_{\text{isog}(2^n, g, 8)}^+$ are equivalent.*

Proof. One direction is given by Lemma 5, and the other by a special case of the discussion above. $\square$

Remark 5. For some practical applications, the proof Π may be replaced by simpler verification procedures. One useful, practical example is the following: if an isogeny of elliptic curves is embedded in a 2-dimensional isogeny, it is easy to see (using the product polarization) that checking that the chain glues is sufficient to verify that the embedded isogeny is non-zero. If one is willing to leak the first steps from the first theta null point (see Section 4.5), and the application allows for an embedding which glues in the first two steps, then this adds nothing to the proof itself, while the verifier need only confirm this gluing.

4.3 Equivalence of $\mathcal{R}_{\text{isog}(2^n, g)}$ and $\mathcal{R}_{\text{HS}(n, g)}$ for $g = 1, 2$

A polynomial time algorithm turning a witness for the statement (A_0, A_1) in $\mathcal{R}_{\text{isog}(2^n, g)}$ into a witness for (A_0, A_1) in $\mathcal{R}_{\text{HS}(n, g)}$ (provided the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion is accessible, see Remark 2) was described in Section 3.1. However, as we saw in Section 3.3, going the other way is more subtle. A witness $(T_i)_{i \in [n-1]}$ for $\mathcal{R}_{\text{HS}(n, g)}$ composed of a path of theta null points of abelian varieties can be used to evaluate the corresponding isogeny $\Phi : A_0 \rightarrow A_1$. Indeed at each step the theta null points of the codomain contain all the information needed to evaluate Φ (see [62, Chap. 8, Sec. 12]; more concretely, [31, Alg. 6] in dimension 2 and [27, Alg. 1] in dimension 4). In dimension 1 and 2 Lemma 3 guarantees that $(T_i)_{i \in [n-1]}$ will always encode an isogeny path. This is enough to prove the following partial result (note that the requirement on the torsion is only needed in one direction, in special cases).

Lemma 7. *Let $g \leq 2$. The relations $\mathcal{R}_{\text{isog}(2^n, g)}$ and $\mathcal{R}_{\text{HS}(n, g)}$ are equivalent, for statements where the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion is accessible.*

In order to say something about the relation to $\mathcal{R}_{\text{isog}(g)}$, we again define the analogous

$$\mathcal{S}_{\text{HS}(n, g, d)} := \left\{ \left(\begin{array}{c} (A_0, A_1), \\ (T_i)_{i \in [n-1]} \end{array} \right) \mid \begin{array}{c} A_0, A_1 \text{ abelian varieties of dimension } g, \\ (((A_0 \times B_0, \theta_0), (A_1 \times B_1, \theta_1)), (T_i)_{i \in [n-1]}) \in \mathcal{R}_{\text{HS}(n, dg)}, \\ \text{for some PPAVs } B_0, B_1, \text{ and level-2 theta structures } \theta_0, \theta_1 \end{array} \right\}.$$

Lemma 8. *There exists a polynomial time algorithm that converts a witness ϕ for the statement (E_0, E_1) in $\mathcal{S}_{\text{isog}(2^n, 1, 2)}$ to a witness Φ for the same statement in the relation $\mathcal{R}_{\text{isog}(1) \leq 2^n}$.*

Proof. The equivalence from Lemma 7 extends to an equivalence between $\mathcal{S}_{\text{isog}(2^n, 1, 2)}$ and $\mathcal{S}_{\text{HS}(n, 1, 2)}$ in the obvious way (and does not require the condition on the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion in this direction). The result now follows from Lemma 6. $\square$

4.4 Weak equivalence of $\mathcal{R}_{\text{isog}(2^n, g)}$ and $\mathcal{R}_{\text{HS}(n, g)}$ for $g > 2$

In dimension $g \geq 3$ the lack of direct correspondence between solutions to Equation (5) and isogenies prevents Lemma 7 from generalizing directly. A solution often adopted for instance in lattice-based cryptography (see e.g. [53, Def. 2.7]) is

to define an alternative relation $\tilde{\mathcal{R}}_{\text{HS}(n,g)}$ where the transcript $\mathcal{T}$ is forced to correspond to an isogeny. We then say that $\mathcal{R}_{\text{isog}(2^n,g)}$ and $\mathcal{R}_{\text{HS}(n,g)}$ are *weakly equivalent* if $\mathcal{R}_{\text{isog}(2^n,g)}$ is equivalent to $\tilde{\mathcal{R}}_{\text{HS}(n,g)}$ (which is now implied by Section 4.3) and for a given statement (A_0, A_1) finding a witness in $\mathcal{R}_{\text{HS}(n,g)} \setminus \tilde{\mathcal{R}}_{\text{HS}(n,g)}$ is hard regardless of the knowledge of a witness in $\tilde{\mathcal{R}}_{\text{HS}(n,g)}$. In fact, we will argue something even stronger: that finding any element at all in $\mathcal{R}_{\text{HS}(n,g)} \setminus \tilde{\mathcal{R}}_{\text{HS}(n,g)}$ is hard, i.e. that Problem 2 is a hard problem.

Either way, the assumption that Problem 1 is hard is exactly saying that $\mathcal{R}_{\text{HS}(n,g)}$ is a hard relation. Problem 2 results in the following lemma.

Lemma 9. *Assuming the hardness of Problem 2, the relations $\mathcal{R}_{\text{isog}(2^n,g)}$ and $\mathcal{R}_{\text{HS}(n,g)}$ (resp. $\mathcal{S}_{\text{isog}(2^n,g,d)}$ and $\mathcal{S}_{\text{HS}(n,g,d)}$) are weakly equivalent for any n, g (resp. any n, g, d), for statements where the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion is accessible.*

Proof. Clear from previous discussion. $\square$

We then finally obtain the following theorem.

Theorem 1. *Assuming the hardness of Problem 2, the relations $\mathcal{R}_{\text{isog}(g) \leq 2^n}$ and $\mathcal{S}_{\text{HS}(n,g,8)}$ are weakly equivalent, for statements where the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion is accessible.*

Proof. Combine Lemma 6 and Lemma 9. $\square$

We again stress that the requirement on the $2^{\lfloor \frac{n}{2} \rfloor}$ -torsion is only necessary for one direction of the equivalence above.

4.5 Change-of-theta coordinates and splitting

A good 2^n -isogeny $(A, \theta^A) \rightarrow (B, \theta^B)$, where the theta-structures are compatible with the isogeny, leaks information about the first and last two steps in the isogeny-chain, by the canonical decomposition of the theta-structures. For many applications, such a leakage would be harmless, but in other cases it may be necessary to randomize the theta-structures at the beginning and end of the chain, by applying a symplectic transformation, changing to a random theta-structure that is independent of the chain. This symplectic transformation must then be included in the zero-knowledge proof. Finally, in the case that a splitting needs to be evaluated (as in Section 4.2), a change of theta-structures at the end will be necessary. In these applications, the change may be chosen to be public (allowing the 2-bit leakage), or private. For these purposes, we now detail how such a change of theta-structure can be checked in zero-knowledge.

Let $M \in \text{Sp}_{2g}(\mathbb{Z}/4\mathbb{Z})$ denote a symplectic basis change, and let the induced theta transformation be denoted by $N \in \text{PGL}_{2g}(\mathbb{F}_{p^2})$, which is computed directly from M using the formula of Robert [62, Chap. 7, App. B]. Proving that $\theta'(P) = N \circ \theta(P)$ is simply a linear relation. Further, we must additionally show that M is

symplectic and that N is the transformation induced by M . Writing $M = \begin{pmatrix} A & C \\ B & D \end{pmatrix}$, the former can be checked using

$$B^T A = A^T B, \quad C^T D = D^T C, \quad A^T D - B^T C = I_g \pmod{4},$$

while the latter can be checked from the aforementioned formula by Robert. If M (and hence N) is allowed to be public, then this can simply be verified on public data, but otherwise, this must also be checked in zero-knowledge.

4.6 Cryptanalysis of Problem 2

As we have seen, if we want $\mathcal{R}_{\text{HS}(n,g)}$ to be a hard relation, we rely on the hardness of Problem 1. Further, if we wish for proving statements in $\mathcal{R}_{\text{HS}(n,g)}$ to be sufficient for proving statements in $\mathcal{R}_{\text{isog}(g)}$, we rely on the hardness of Problem 2.

To the best of our knowledge, neither problem has been studied before. We will in this section limit our discussion to Problem 2, since the hardness of Problem 2 already implies that the usual isogeny-path problem reduces to Problem 1.

If we fix a certain theta null point X^0 in Equation (5) we see that the possible solutions are given by

$$X^1 = \frac{1}{2^g} \mathcal{H} \circ \mathcal{T}_s \circ \mathcal{H} \circ \mathcal{S}(X^0) \tag{9}$$

where

$$\mathcal{T}_s : \mathbb{P}^{2^g-1} \rightarrow \mathbb{P}^{2^g-1} : (x_0 : \dots : x_{2^g-1}) \rightarrow (x_0 : s_1 \sqrt{x_0 x_1} : \dots : s_{2^g-1} \sqrt{x_0 x_{2^g-1}})$$

is the map defined in [48, Cor. 5] with the difference that we now allow all $2^g - 1$ possible sign choices. If we at any point take an “invalid” choice of square roots, we are likely to step out of the moduli space defined by the theta null points of abelian varieties. Heuristically, if we then model the resulting X^1 as a random element of $\mathbb{P}^{2^g-1}$, there is a priori no reason why the map $\mathcal{T}_s$ should produce new elements in $\mathbb{P}^{2^g-1}(k)$. This happens if and only if all the square-roots are rational, which happens with probability $1/2^{2^g-1}$ when working over finite fields, thus most “invalid” path choices end immediately. We illustrate this in Figure 1 in Appendix B, where we explore the neighbourhood of a theta null point coming from a superspecial abelian threefold.

Even ignoring the fact that most counterfeit paths end immediately, the ability to produce random values would cryptographically not be enough to break Problem 2. Under the heuristic that the pseudo-path null points act as random points of $\mathbb{P}^{2^g-1}$, the moduli space defined by the theta null points of abelian varieties of dimension $g > 2$ always has co-dimension ≥ 1 , and thus we roughly expect such a random, brute-force strategy to have complexity $O(p)$ in the absolute best case of $g = 3$ (and typically, much worse). We confirm this experimentally in Appendix B: a depth-first search for solutions to Problem 2 over superspecial abelian threefolds shows the complexity of such a brute-force

search growing very quickly with the characteristic p , and a breadth-first search around singular theta null points yields no solutions either. The code used for our tests can be found in our repository.

Remark 6. Given the novelty of Problem 2, we cannot exclude extra structure in these pseudo-isogeny graphs that could be exploited. We note a vague connection to the long-standing open problem of hashing into the supersingular isogeny graph in dimension 1 (see [13]): since solutions to Problem 2 need not correspond to isogenies, such a solution could plausibly yield a higher-dimensional analogue of that hash, though this connection remains speculative.

5 Constraint systems for proofs of knowledge of isogenies

In the previous sections, we have shown that proving knowledge of a length n path of 2-isogenies between two principally polarized abelian varieties A_0, A_n of dimension g is computationally equivalent to proving knowledge of a vector $v \in (k^{2^g})^{n+1}$ such that $v[0] = \theta_{A_0}, v[n] = \theta_{A_n}$, and

$$\forall i \in [n-1] : \mathcal{S} \circ \mathcal{H}(v[i+1]) = \mathcal{H} \circ \mathcal{S}(v[i]). \quad (10)$$

We now transform the relation in Equation (10) to an equivalent relation that is more conducive to the construction of a zero-knowledge proof of knowledge. Define the Hadamard matrix $H \in k^{2^g \times 2^g}$ such that $\mathcal{H} : k^{2^g} \rightarrow k^{2^g} : x \mapsto Hx$. We prove knowledge of a vector $z \in k^{2^{g+1}(n+1)}$ such that $z[0 \dots 2^g - 1] = \theta_{A_0} \wedge z[2^g n \dots 2^g(n+1) - 1] = \theta_{A_n}$ and

$$\left(\left[\begin{array}{c|c} I & 0 \\ \hline 0 & H \ 0 \ 0 \ 0 \\ 0 & 0 \ \ddots \ 0 \ 0 \\ 0 & 0 \ 0 \ H \end{array} \right] z \right)^2 = \left[\begin{array}{c|c} 0 & I \\ \hline 0 & H \ 0 \ 0 \ 0 \\ 0 & 0 \ \ddots \ 0 \ 0 \\ 0 & 0 \ 0 \ H \ 0 \end{array} \right] z, \quad (11)$$

where the square of a vector denotes the Hadamard product with itself. Clearly, $z = (v, v \circ v)$, with v the vector containing the theta null points of the isogeny path, satisfies this relation. In the following we will denote the matrix on the left by M_1 and the one on the right by M_2 , so the above constraint system can be written as $(M_1 z) \circ (M_1 z) = M_2 z$.

Remark 7. Since Equation (7) closely resembles Equation (10), the techniques in this section trivially extend to additionally prove the finiteness of the relevant kernel, as discussed in Section 4.2. For the purpose of brevity, we do not discuss this extension here. However, the constraint system parameters in Table 1 and implementation results for $g = 4$ in Section 8 all include these extra relations.

Optimizing constraint systems. Previous research on proving knowledge of isogeny relations often follows the strategy: 1) construct an alternative relation that is equivalent to the isogeny relation, 2) express this alternative relation in a constraint system for which there exist post-quantum secure proof systems, 3) gauge the performance of the resulting proof system based on the size of the resulting constraint system. Usually, the relations are expressed in the R1CS constraint system which is defined by three sparse matrices $A, B, C \in k^{m \times n}$ with $\mathcal{O}(m) = \mathcal{O}(n)$ non-zero entries such that a transcript $z \in k^n$ satisfies this constraint system if and only if $(Az) \circ (Bz) = Cz$.

As a result, previous research has focused mostly on minimizing the dimensions and the number of non-zero entries when expressing the isogeny relation in an R1CS constraint system. Research on proof systems has conclusively shown that, even though this cannot give any asymptotic improvement in performance, practical performance can be greatly improved by expressing the relation in specialized constraint systems such as Plonk, CCS and GKR. Let us illustrate this with a simple example. The relation in Equation (11) can be expressed in R1CS by setting $A = B = M_1$ and $C = M_2$. However, it should be clear that the performance of any proof system for R1CS can be improved by specializing it to R1CS instances where $A = B$. Additionally, many of the most efficient proof systems will actually pad the number of rows and columns to a power of two and thus have limited performance difference within the same power of two.

For these reasons, we will also report actual concrete implementation results in the following sections, whereas most previous works relied on extrapolated estimates.

Representing $\mathbb{F}_{p^2}$ as $\mathbb{F}_p \times \mathbb{F}_p$. In the most important use case of dimension 4, namely the CSIDH class group action, we will have $k = \mathbb{F}_p$ for some prime p . In dimension 1 and 2 however, most use cases will have $k = \mathbb{F}_{p^2}$. Due to many proof system implementations only natively supporting prime fields, previous work has discussed constraint system sizes resulting from both expression over the native field $\mathbb{F}_{p^2}$, as well as expression over $\mathbb{F}_p$ by emulating $\mathbb{F}_{p^2}$. We will also discuss both approaches using the same techniques for representing $\mathbb{F}_{p^2}$ as $\mathbb{F}_p \times \mathbb{F}_p$ and report implementation results comparing both.

Note that any element in $\mathbb{F}_{p^2}$ can be written as $a + b\alpha$ where $\alpha^2 = d$ is a non-residue. To express the square $e + f\alpha := (a + b\alpha)^2$ using the R1CS constraint system, we append $\mathbb{F}_p$ elements $u := (a + b)(a + bd)$ and $w := ab$ to the transcript. Expressing the relation between a, b, u, w can be achieved using only two R1CS constraints. The $\mathbb{F}_p$ elements representing the square can now be expressed “for free” as $e = u - (d + 1)w$ and $f = 2w$.

Recall the matrices $M_1, M_2 \in k^{(2^{g+1}(n+1)-2^g) \times 2^{g+1}(n+1)}$ such that Equation (11) is equivalent to $(M_1 z) \circ (M_1 z) = M_2 z$. We use this constraint system to prove isogeny relations in dimension $g = 4$ for $k = \mathbb{F}_p$ and in dimensions $g = 1, 2$ for $k = \mathbb{F}_{p^2}$. Let us briefly describe our approach for proving isogeny relations in dimension $g = 1$ using proof systems over $k = \mathbb{F}_p$. We write one step

of Equation (10) for $v[0] = [x_1 + y_1\alpha, x_2 + y_2\alpha]$ and $v[1] = [x_1 + y_1\alpha, x_2 + y_2\alpha]$ as

$$\begin{aligned} x_1^2 + y_1^2d \pm (x_2^2 + y_2^2d) &= x_1^2 + y_1^2d \pm 2(x_1x_2 + y_1y_2d) + x_2^2 + y_2^2d \\ 2x_1y_1 \pm 2x_2y_2 &= 2x_1y_1 \pm 2(x_1y_2 + x_2y_1) + 2x_2y_2, \end{aligned}$$

which we can rewrite (using the u and w notation introduced above) as

$$\begin{aligned} u_1 - (d+1)w_1 \pm u_2 - (d+1)w_2 &= u_1 - (d+1)w_1 \pm 2(x_1x_2 + y_1y_2d) + u_2 - (d+1)w_2 \\ w_1 \pm w_2 &= w_1 \pm 2(x_1y_2 + x_2y_1) + w_2. \end{aligned}$$

Notice that we cannot express these four equations as only four rows of an R1CS instance. Instead, we add the cross-terms x_1x_2, x_1y_2 to the transcript (which each require one R1CS row to express), and then express the previous equations as four R1CS constraints. We follow a similar approach for dimension $g = 2$. In this case, the square of the Hadamard transform results in six cross-terms and similarly we add two finite field elements per cross-term. We refer to Table 1 for total sizes of the resulting R1CS constraint systems.

	$g = 1, k = 256$			$g = 2, k = 108$			$g = 4, k = 498$		
	m	n	nz	m	n	nz	m	n	nz
$\mathbb{F}_p$	2556	2560	12772	4096	4104	28560	31874	31872	811206
$\mathbb{F}_{p^2}$	1024	1024	3064	864	864	4288	n/a		

Table 1: The number of rows m , number of columns n and the number of non-zero elements of the R1CS constraint system resulting from expressing Equation (10) in R1CS over either $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ for dimensions $g = 1, 2, 4$ where k is the path length corresponding to the applications described in Section 7. In the cases that do not require emulation, we do not count non-zero elements of matrix B since it is exactly equal to A .

6 Proof systems for proving knowledge of isogenies

In this section we will discuss different proof systems that prove the constraint systems described in Section 5. Since we argue that proof systems for proving knowledge of isogeny relations should ideally be compared based on measured runtimes instead of sizes of R1CS constraint systems, we provide concrete runtimes in Section 8.

Furthermore, we want to point out that all succinct non-interactive proof systems must rely on non-falsifiable computational assumptions [41]. Since no succinct isogeny-based proof system is currently known in the literature, proof systems for proving knowledge of isogeny relations must make additional assumptions besides the isogeny path problem. Ideally, these proof systems only

additionally assume the existence of (quantum) random oracles, which is already common in isogeny-based cryptography. For this reason, previous work refers to hash-based succinct proof systems such as Aurora [9] and Ligero [3] whose soundness relies on the proximity gaps phenomenon for Reed-Solomon codes. However, the most efficient instantiations of hash-based proof systems (in terms of proof size and verification time) rely on parameter sets for the proximity gaps that were only conjectured to be secure. Recent work has now disproven this conjecture [37, 26]. Instantiating these proof systems with provably secure parameters will increase proof size and verification time by at least a factor of 2–3.

For this reason, we will describe both a hash-based proof system and a lattice-based proof system that is able to achieve smaller proof size and verification time at the cost of making additional assumptions. In Section 6.1, we describe a multilinear proof system that uses the BaseFold [66] hash-based polynomial commitment scheme as a building block. We choose BaseFold since it is completely field-agnostic. Note that many efficient proof systems will require that $\mathbb{F}_p$ contains 2^k -th roots of unity with $k \gg 0$, which is incompatible with the $p \equiv 3 \pmod{4}$ requirement common for many isogeny use cases. In Section 6.2, we design a lattice-based proof system that achieves a lower proving time and a smaller proof size.

6.1 Multilinear PIOP for Equation (11)

This section describes a Polynomial Interactive Oracle Proof (PIOP) [10] for the relation implicit in Equation (11). This is an adaptation of generic techniques for multilinear PIOPs [64] to our constraint system. As discussed in Section 5, emulating $\mathbb{F}_{p^2}$ for dimensions $g = 1, 2$ will instead require PIOPs for the full R1CS constraint system.

Let us define the transcript size $N := 2^{g+1}(n+1)$ and $m := \log_2(N)$. For ease of notation we assume M_1, M_2 were padded to become square matrices. Our proof of knowledge starts by proving knowledge of vectors $z_1, z_2 \in k^N$ such that $z_1 \circ z_1 = z_2$. We start by restating this relation using multilinear polynomials

$$\begin{aligned} z_1^2 = z_2 &\Leftrightarrow \forall X \in \{0, 1\}^m : \tilde{z}_1(X)^2 = \tilde{z}_2(X) \\ &\Leftrightarrow F_{rc}(X) := \sum_{b \in \{0, 1\}^m} \tilde{\text{eq}}(X, b)(\tilde{z}_1(b)^2 - \tilde{z}_2(b)) = 0. \end{aligned}$$

For some random $\tau \in k^m$, the Schwartz-Zippel lemma (Lemma 4) states that $F_{rc}(\tau) = 0$ implies $F_{rc} = 0$ except with probability $\mathcal{O}(1/|k|)$. The multilinear sumcheck protocol reduces this sum to the claim that

$$h_m(r_m) = \tilde{\text{eq}}(\tau, r)(\tilde{z}_1(r)^2 - \tilde{z}_2(r)) \quad (12)$$

where $h_m \in k^{\leq 3[Y]}$ is the last polynomial sent in the sumcheck protocol and $r \in k^m$ is the random evaluation point. The verifier checks the claim $h_m(r_m) \stackrel{?}{=} \tilde{\text{eq}}(\tau, r)(a_1^2 - a_2)$ where $a_1, a_2 \in k$ were provided by the prover and the evaluations

$\tilde{eq}(\tau, r), h_m(r_m)$ can be computed by the verifier themselves. Now the verifier only has to check the validity of the claim that $a_1 \stackrel{?}{=} \tilde{z}_1(r)$ and $a_2 \stackrel{?}{=} \tilde{z}_2(r)$.

Again, we restate this claim using multilinear polynomials so that we can leverage the sumcheck protocol. For some $i \in \{1, 2\}$ and $\tilde{z}$ a multilinear extension of z , we can see that

$$a_i = \tilde{z}_i(r) \Leftrightarrow a_i = \sum_{b \in \{0,1\}^m} \tilde{M}_i(r, b) \tilde{z}(b)$$

where $\tilde{M}_i$ is a multilinear extension of the M_i matrix such that the first m variables represent the row index and the last m variables represent the column index. Again, we can leverage the Schwartz-Zippel lemma (Lemma 4) to check this claim for all i by instead verifying whether $r_1^* a_1 + r_2^* a_2 \stackrel{?}{=} r_1^* \tilde{z}_1(r) + r_2^* \tilde{z}_2(r)$. The sumcheck protocol will reduce this claim to

$$h'_m(r'_m) \stackrel{?}{=} (r_1^* \tilde{M}_1(r, r') + r_2^* \tilde{M}_2(r, r')) \cdot \tilde{z}(r').$$

Notice that $\tilde{M}_1(r, r'), \tilde{M}_2(r, r')$ can be efficiently computed by the verifier themselves, but $\tilde{z}(r')$ has to be provided by the prover in a verifiable way. We refer to Figure 2 in Appendix C for a formal description of the multilinear PIOP described in this section.

This PIOP can be compiled into a zero-knowledge Succinct Non-interactive ARgument of Knowledge (zkSNARK) by applying standard techniques [10, 64]. As previously mentioned, our implementation will prove the final evaluation of $\tilde{z}$ using the BaseFold Polynomial Commitment Scheme (PCS). Adding zero-knowledge to this PIOP and making it non-interactive will not be discussed since it has negligible impact on performance.

6.2 Lattice-based sigma protocol with linear and quadratic relations

In this section, we propose a new type of proof system based on lattice hardness assumptions instead of the proximity gaps conjecture. Importantly, this proof system should also be post-quantum secure, field-agnostic and zero-knowledge. Relative to the proof system described in Section 6.1, we achieve both lower prover computation costs and lower proof size. Note that, even though it is asymptotically succinct, the previous proof system has proof size of 9MiB in dimension 4 for a transcript that has size 2MiB. This can be ascribed to the fact that BaseFold is not practically succinct for small instances. The proof system below will achieve a 2.7MiB proof size while being approximately twice as fast. In particular, we propose a lattice-based sigma protocol that proves Equation (10) by recursively proving knowledge of a verifiable proof for the PIOP described in Section 6.1. We first describe the lattice-based sigma protocol and then describe how it can prove Equation (10).

The protocol from Figure 3 in Appendix C proves knowledge of an opening $c \in k^{m_2}$ for a BDLOP [8]-style commitment Com constructed as

$$\begin{bmatrix} t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \cdot s + \begin{bmatrix} 0 \\ c \end{bmatrix} \quad (13)$$

for some public parameters $B_1 \in k^{m_1 \times r}$, $B_2 \in k^{m_2 \times r}$ and some secret $s \in k^r$ such that $\|s\| \leq B_s$. We define the challenge distribution $\Gamma := \{\gamma \mid \gamma \leq B_\gamma\} \subset k$ and the discrete Gaussian distribution $\mathcal{D}_\sigma$ over k with standard deviation σ . Notice that this commitment scheme is binding if $\text{SIS}_{k,m_1,r,2B_s}$ is hard. Similarly, the hiding property depends on the hardness of $\text{LWE}_{k,r-m_1-m_2,m_1+m_2,B_s}$. This sigma protocol is able to prove v linear relations expressed as $Rc = u$ using the matrix $R \in k^{v \times m_2}$ and $u \in k^v$.

The sigma protocol from Figure 3 derives its simulatability from the hiding property of the commitment scheme combined with the rejection sampling, which ensures that z is statistically indistinguishable from a random value sampled from $\mathcal{D}$. Let us discuss the knowledge soundness property in more detail. Notice that an extractor can perform rewinding to obtain two accepting transcripts $(w, v, \gamma^{(1)}, z^{(1)}), (w, v, \gamma^{(2)}, z^{(2)})$ such that

$$B_1(z^{(1)} - z^{(2)}) = t_1(\gamma^{(1)} - \gamma^{(2)}) \quad (14)$$

and thereby effectively extracting an opening

$$\bar{s} := \bar{z}/\bar{\gamma} := (z^{(1)} - z^{(2)})/(\gamma^{(1)} - \gamma^{(2)}).$$

It should be clear that the opening also satisfies $R\bar{c} = u$ for $\bar{c} := t_2 - B_2\bar{s}$. Next, we show that $\bar{s}$ (and therefore also $\bar{c}$) is unique given the hardness of $\text{SIS}_{k,m_1,r,8B_s\sigma\sqrt{2r}}$. Any extracted second opening $\bar{s}'$ has to also satisfy Equation (14) and thus

$$B_1(\bar{s} - \bar{s}') = 0 \Leftrightarrow B_1(\bar{s}\bar{\gamma}\bar{\gamma}' - \bar{s}'\bar{\gamma}\bar{\gamma}') = 0 \Leftrightarrow B_1(\bar{z}\bar{\gamma}' - \bar{z}'\bar{\gamma}) = 0 \quad (15)$$

for $\|\bar{z}\bar{\gamma}'\|, \|\bar{z}'\bar{\gamma}\| \leq 4B_\gamma\sigma\sqrt{2r}$. Lastly, we show this protocol is complete except with negligible probability. Clearly, the two linear relations checked by the verifier will always be satisfied for honest provers. Following [52, Lemma 2.2], we can state that the inequality will hold with overwhelming probability as long as $r \geq 640$. Following [52, Lemma 2.14], the rejection sampling subroutine returns after an expected $M \approx 3$ repetitions if we set $\sigma = 13B_\gamma B_s$. The proof size of this protocol is approximately $|t| + |z| + |\gamma|$ in the non-interactive setting. In that case, the verifier uses the linear relations to compute w, v and then uses them to audit the sampling of γ through Fiat-Shamir with aborts [51].

Notice that almost all verification steps in the protocol from Section 6.1 are linear and can therefore be expressed as a linear relation $Rc = u$ where c contains the proof transcript. The only exception is Equation (12), which we cannot prove using the protocol from Figure 3. We can restate this as proving knowledge of h_m, a_1, a'_1, a_2 such that $a'_1 = a_2 + h_m(r_m)/\tilde{e}\tilde{q}(\tau, r)$ and $a'_1 = a_1^2$. To achieve this we define the Laurent polynomials

$$\begin{aligned} p(Y) &:= a_1(Y + Y^{-1}) + a'_1 Y^2 + a_3 Y^3 \\ q(Y) &:= p(Y) \cdot (p(Y) - 2Y^{-2}) \end{aligned}$$

for some random $a_3 \in k$. Now notice that the constant coefficient of q vanishes if and only if $a'_1 = a_1^2$. First, the prover commits to a_1, a'_1, a_3, a_2 and to $q(Y) =$

$q_1Y^{-3} + q_2Y^{-2} + q_3Y^{-1} + q_4Y + q_5Y^2 + q_6Y^3 + q_7Y^4 + q_8Y^5 + q_9Y^6$ whose constant coefficient is zero by construction. Then, the verifier responds with a randomly sampled $y \leftarrow k$. The prover responds with $p(y), q(y)$, which the verifier can use to check $q(y) \stackrel{?}{=} p(y)(p(y) - 2y^{-2})$. Since only one evaluation of $p(Y)$ will be known to the verifier, the addition of a_3 is sufficient for zero-knowledge.

Proof composition. In the interest of minimizing the proof size, we propose performing one recursive step. In other words, we recursively prove knowledge of a valid proof for the proof system described in Figure 2 using the proof system described in Figure 3. Specifically, the prover will simulate execution of the multilinear PIOP to generate a transcript c , which will become the message committed to in Equation (13). Then the prover executes the “outer” proof system, i.e. the one described in Figure 3.

Notice that this proof system is capable of efficiently representing all verification operations of the “inner” proof system except the Fiat-Shamir transform required to make it non-interactive. We can avoid proving the Fiat-Shamir transform in our outer proof system by first transforming the inner proof system using the commit-and-prove transform before proof composition. Concretely, every prover message in Figure 2 is replaced by a commitment to that message. The verifier does not perform any operation on the messages and only sends back random challenges. As a final message, the prover opens all those commitments. Then, the verifier can verify those openings and perform the deferred verification operations.

To summarize, we transform the protocol from Figure 2 to a commit-and-prove protocol that uses the lattice-based commitment scheme from Equation (13). Next, we use the Fiat-Shamir transform to make the proof system non-interactive. Lastly, we prove all verification operations on commitment openings in the resulting protocol recursively using the proof system from Figure 3. We present a formal description of the interactive variant (before the Fiat-Shamir transform) of the resulting protocol in Figure 4 in Appendix C.

7 Applications

7.1 Proofs of isogeny relations

Our main theoretical contribution is providing a zero-knowledge proof of knowledge for the relation

$$\mathcal{R}_{\text{isog}(g)} := \left\{ ((A_0, A_1), \phi) \mid \begin{array}{l} A_0, A_1 \text{ abelian varieties of dimension } g, \\ \phi: A_0 \rightarrow A_1 \text{ is an arbitrary } N\text{-isogeny for some } N \in \mathbb{N} \end{array} \right\}.$$

This fully solves (a generalization of) one of the major open problems stated in [11], namely how to prove $\mathcal{R}_{\text{isog}}$ without knowledge of endomorphism rings and without leaking the degree of ϕ . As already observed there, tailored protocols like SQIsign rely on nontrivial information about the curves involved (e.g. their

endomorphism rings) while generic proofs in dimension 1 are constrained to $\deg(\phi)$ being smooth (on top of leaking $\deg(\phi)$ itself).

Another important relation mentioned in [11] is for CSIDH isogenies

$$\mathcal{R}_{\text{CSIDH}} := \{((E_0, E_1), \phi) \mid \phi : E_0 \rightarrow E_1 \text{ is an isogeny defined over } \mathbb{F}_p\}.$$

Thanks to **qt-Pegasis**, such an isogeny ϕ can always be embedded into a 2^n -isogeny in dimension 4, i.e. $\mathcal{R}_{\text{CSIDH}}$ can be expressed in $\mathcal{R}_{\text{isog}(2^n, 4)}$. Our framework thus applies directly, providing a completely novel proof of knowledge for $\mathcal{R}_{\text{CSIDH}}$ compared to the graph isomorphism protocol due to Couveignes. This shows once more the great generality of our approach. The fact that, as discussed in Section 8, our proving time is comparable to the ID-protocol by Couveignes using the recent heavily optimized implementation of **qt-Pegasis** [28] (although proof sizes are 1 to 2 orders of magnitude larger, see Section 8) furthermore shows that this flexibility comes at a moderate cost.

Remark 8. When the secret isogeny is embedded using ad-hoc techniques like **qt-Pegasis** rather than Lemma 2, care is required to ensure the embedding remains zero-knowledge for every secret isogeny, i.e. that any secret isogeny can be embedded using such a technique. For $\mathcal{R}_{\text{CSIDH}}$ this amounts to proving that the failure probability of **qt-Pegasis** is negligible for the chosen parameters.

7.2 Additional isogeny relations

Given the generality of our framework, extending it to prove more specific isogeny relations is natural. We leave a detailed treatment for future work, but sketch two examples below.

Proving the evaluation of points, as we have documented and implemented, is conceptually very similar to proving a path of theta null points, and may have further applications beyond those discussed here (see e.g. [4, Sec 5.2]).

Another property worth proving is the degree of an isogeny, captured by the relation $\mathcal{R}_{\text{deg}}$ in [11] and used e.g. by PRISM [5]. Since Lemma 2 hides the degree of the embedded isogeny, adding a degree proof amounts to performing PRISM’s verification in zero-knowledge, which requires evaluating points (discussed above) and a pairing computation, a well-studied problem [39, 57] that we expect to adapt to the level 2 theta model.

Together, proving evaluations of points and degrees already implies a proof for $\mathcal{R}_{\text{M-SIDH}}$, the only relation from [11] not explicitly covered above.

7.3 Applications to higher-dimensional isogenies

As a more concrete application of our construction we already mentioned an alternative proof of knowledge of a CSIDH isogeny built with **qt-Pegasis**. Another interesting application is a trusted setup for the 2-dimensional CGL hash function [17, 48]. In a similar fashion to how KLPT [47] showed that the original CGL hash function [20] is insecure if instantiated from a curve of known

endomorphism ring, the recent 2-dimensional variant of KLPT [16] showed that the same is true for the 2-dimensional variant of CGL. The trusted setup for CGL proposed in [6] obviates this problem in dimension 1, and is currently one of the main applications of generic proofs of knowledge of isogenies. Thanks to our construction, this multiparty trusted setup can be replicated directly in dimension 2. We provide a proof-of-concept implementation proving a random walk from [48]. To the best of our knowledge, this is the first instantiation of the 2-dimensional CGL hash function which is secure against [16]. Notice that, since we are working in dimension 2, we do not need to rely on the hardness of Problem 2.

Finally, recent work by Robert [63] constructs a non-interactive key exchange, $\otimes$ -MIKE, using 2^n -isogenies in dimension 4; our framework applies directly to proving such an isogeny, and may prove useful in future protocols built on $\otimes$ -MIKE.

8 Implementation

We implement our construction in the following cases:

- in dimension 1 and 2, we prove knowledge of a random walk; as discussed in Section 7.3, in dimension 2 this can be used to securely instantiate the dimension 2 CGL hash function, preventing the attack from [16];
- in dimension 4, we prove the computation of a **qt-Pegasis** isogeny [30]; this provides an alternative proof of the $\mathcal{R}_{\text{CSIDH}}$ relation discussed in Section 7.1.

We could not implement our construction in dimension 8 since, to the best of our knowledge, 8-dimensional isogenies in the level-2 theta model have not been implemented yet. However, we stress that once such an implementation exists, adapting it to also produce transcripts for our framework will not be more difficult than in dimension 2 or 4. Our implementation can be found in our repository.

8.1 Building the transcript

We implement the generation of a transcript satisfying Equation (5) directly on top of some of the most common isogeny computation libraries, to facilitate the usability of our code. In particular, we make the following choices:

- in dimension 1, we build on the implementation of [31]; we work over $\mathbb{F}_{p^2}$ where $p = 5 \cdot 2^{248} - 1$ is the SQIsign level I prime, and perform a random walk of length 256 from $E_0 : y^2 = x^3 + x$;
- in dimension 2, we prove a random walk as an evaluation of the CGL hash function from [48] (originally due to [65]); we adopt their parameters, i.e. $\mathbb{F}_{p^2}$ where $p = 2^{127} - 1$, and perform a random walk of length 108 from $E_0 \times E_0$;
- in dimension 4, we integrate the transcript generation into the **qt-Pegasis** code; we thus work over $\mathbb{F}_p$ where $p = 27 \cdot 2^{500} - 1$ and perform 498 steps.

The procedure is the same in all cases: for each isogeny step we can use the theta coordinates X^j in the transcript, and then update the coordinates of X^{j+1} so that Equation (4) holds affinely. Since it already holds projectively, it is enough to compute for instance the coefficient c_j such that

$$\sum_{i=0}^{2^g-1} (X_i^j)^2 = c_j \left(\sum_{i=0}^{2^g-1} X_i^{j+1} \right)^2,$$

and then divide X^{j+1} by $\sqrt{c_j}$. This costs a single square root per step. Alternatively, one could also keep track of the coefficients c_j and modify the proof system accordingly. Since all the above mentioned libraries work projectively, it is always possible to scale X^j by a given factor. Notice that this procedure works for all types of isogenies (e.g. also for gluing and splitting isogenies).

8.2 Proof systems

In Table 2, we report measured results from implementing the BaseFold-based proof system described in Section 6.1 in Rust and running it on an Apple Mac mini M4 Pro with 64GB RAM.

		$g = 1, k = 256$	$g = 2, k = 108$	$g = 4, k = 498$
$\mathbb{F}_p$	Prover time	1288ms	966ms	22475ms
	Proof size	4826KiB	6320KiB	8835KiB
	Verif. time	74ms	45ms	257ms
$\mathbb{F}_{p^2}$	Prover time	1846ms	1074ms	
	Proof size	3344KiB	3767KiB	n/a
	Verif. time	286ms	179ms	

Table 2: The prover time, proof size and verification time over either $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ for dimensions $g = 1, 2, 4$ where k is the path length corresponding to the applications described in Section 7.

We would like to note that using similar extrapolation techniques as in [45], we would arrive at smaller and faster proofs, e.g. 17s prover time and 1MiB proof size for our dimension 4 case using the Aurora proof system. However, as explained before, the parameters used in these implementations are not provably secure. We refer to Table 3 for our instantiations of the lattice parameters described in Section 6.2 such that the SIS and LWE instances mentioned are secure for $\lambda = 128, g = 4$ using the lattice estimator by Albrecht et al. [2].

m_1	m_2	r	B_s	B_γ	p
2^{13}	32887	$2^{14} + 2^{12} + m_1 + m_2$	2^{28}	2^{128}	$27 \cdot 2^{500} - 1$

Table 3: Lattice-based sigma protocol parameters described in Section 6.2.

Measured results from our implementation are reported in Table 4. We would like to note that even though the proof system described in Section 6.2 is based on the LWE problem for simplicity, we instead implemented a version of this proof system extended to $k[X]/(X^{2^d} + 1)$, i.e. the RLWE version, using standard techniques. This results in increased verifier performance (compared to $d = 0$), at the cost of slightly worse prover performance.

d	Prover time (s)	Proof size (KiB)	Verification time (s)
5	11.5	2747	299
7	12.9	2759	79.0
9	14.4	2931	22.9
11	16.1	3120	9.3

Table 4: Results for proof system described in Section 6.2 for $g = 4$ and $k = 498$.

In dimension 1, our R1CS instance is roughly a factor of 2 bigger than the state of the art, since we use 2 projective coordinates instead of a single affine one; this choice illustrates the generality of our approach, with the dimension 1 instance serving as an introduction to dimensions 2 and 4.

In dimension 4, the tailored CSI-FiSh proof using the optimized qt-Pegasis implementation [28] costs about 3 seconds and an 8.3KiB proof for 128 actions; our fully general approach is less than 4 times slower in proving time, though proofs are 2.7MiB. Both figures are largely inflated by our longer chain over a larger field, so applications not threatened by Kuperberg’s algorithm should instead be extrapolated from our dimension 1 and 2 results.

In dimension 2, our application is fully novel: the timing and sizes in Table 2 estimate the efficiency of the CGL hash trusted setup described in Section 7.3.

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A Additional relations

A.1 Additional relations for $g > 2$

In principle, it is possible to prove that the solutions to Equation (5) actually come from theta null points of principally polarized abelian varieties even for $g > 2$, by providing additional relations. The missing ingredient is that we have to express that each of the 2^g -tuples T_i in the solution actually corresponds to the theta null point of a principally polarized abelian variety.

These extra relations can be derived explicitly from the Riemann relations [56] between level-4 theta null points (see [48, Proposition 7] for this proof technique). From these Riemann relations, we will then derive $(2^g - 1) - (g(g + 1)/2)$ extra conditions that define valid level-2 theta null points.

Dimension $g = 3$. An example of this in dimension $g = 3$ is the following, which is taken directly from [48]. For $g = 3$ we have $2^g - 1 = 7$ and $g(g + 1)/2 = 6$ and we thus need a single additional equation, that can be derived directly from the Riemann relations.

Theorem 2. *Let $T := (a_{000}, \dots, a_{111}) \in \mathbb{A}^8$ (indices are in binary). Defining*

$$(\beta_{00}, \dots, \beta_{11}, \delta_{00}, \dots, \delta_{11}) = \mathcal{H} \circ \mathcal{S}(T)$$

and for $i \in (\mathbb{Z}/2\mathbb{Z})^2$, $\gamma_i := 2 \sum_{j \in (\mathbb{Z}/2\mathbb{Z})^2} (-1)^{\langle i, j \rangle} a_{0j} a_{1j}$,

$$R_1 = \prod_i \beta_i, \quad R_2 = \prod_i \gamma_i, \quad R_3 = \prod_i \delta_i,$$

then $(a_{000} : \dots : a_{111}) \in \mathbb{P}^7$ is the level-2 theta null point of an abelian variety of dimension 3 if and only if

$$R_1^2 + R_2^2 + R_3^2 - 2(R_1 R_2 + R_1 R_3 + R_2 R_3) = 0. \quad (16)$$

Further, this relation is enough to completely determine the last sign in Lemma 3.

Proof. See [48, Proposition 7] and [48, Theorem 8]. □

The above theorem shows that for $g = 3$ it suffices to add one extra non-linear condition given by Equation (16) on T for it to be a valid level-2 theta null point. Note that even in this case the degree of this extra relation is already 8.

Dimension $g = 4$. A similar approach can be taken for dimension $g = 4$. Let $T := (a_{0000}, \dots, a_{1111}) \in \mathbb{A}^{16}$, then in this case we will need to derive $5 = (2^g - 1) - (g(g + 1)/2)$ extra relations amongst the coordinates of T for it to represent a valid level-2 theta null point. We will not derive these equations fully, but only sketch the complexity involved. Following a similar technique as [48, Proposition 7], these extra relations will be derived from the Riemann relations between level-4 theta null coordinates. Compared to a level-2 theta structure, a level-4 theta structure defines a map $\theta^A : A \rightarrow \mathbb{P}^{2^{2g}-1}$ and we can thus index these level-4 coordinates by elements of $(\mathbb{Z}/2\mathbb{Z})^{2g}$.

The Riemann relations (see for instance [40, Equation 2.1] for an explicit expression) define identities between level-4 theta null coordinates of abelian varieties. To illustrate the nature of these relations, we provide an example.

Example 1. Since we have $2^{2g} = 256$ level-4 theta coordinates for $g = 4$, we will index them with integers $0 \leq i < 256$. One of the many Riemann relations that needs to hold is then given by:

$$\begin{aligned} 3 \cdot [0, 64, 128, 192] = & [1, 65, 129, 193] + [2, 66, 130, 194] + [3, 67, 131, 195] + \\ & [16, 80, 144, 208] + [18, 82, 146, 210] + [32, 96, 160, 224] + \\ & [33, 97, 161, 225] + [48, 112, 176, 240] + [51, 115, 179, 243] \end{aligned}$$

In the above, each vector with 4 indices corresponds to a product of the corresponding level-4 theta null coordinates, e.g. $[0, 64, 128, 192] = \theta_0 \cdot \theta_{64} \cdot \theta_{128} \cdot \theta_{192}$. The above relation between level-4 theta null coordinates can then be turned into a relation between level-2 theta null coordinates by using the fact that the squares of the level-4 theta null coordinates can be written as a combination of the level-2 theta null coordinates. If we denote each of the 10 terms in the above equation as r_i for $i = 0, \dots, 9$, then the above equation now reads $3r_0 - \sum_{i=1}^9 r_i = 0$. To derive an equation involving only the squares of the r_i (since these are expressions in the level-2 theta null points), we need to consider the product over all possible sign twists

$$\prod_{(\delta_1, \dots, \delta_9) \in \{0,1\}^9} (3r_0 - \sum_{i=1}^9 (-1)^{\delta_i} r_i) = 0.$$

Note that this equation defines a relation of degree 2^8 between the squares of the r_i . Since each r_i is a product of four level-4 theta null coordinates, and each square of a level-4 theta null coordinate can be written as a linear combination of a product of two level-2 theta null coordinates, we conclude that the above equation defines a relation of degree 2048 between the level-2 theta null coordinates.

□

As the above example illustrates, the extra condition coming from the Riemann relations is given by a polynomial of degree 2^8 in 10 variables, namely the squares r_i^2 , where each of the r_i^2 is an expression of degree 8 in the level-2 theta null coordinates. Although these equations are very explicit, their complexity

is totally prohibitive (the homogeneous polynomial of degree 2^8 in 10 variables would involve $> 2^{53}$ terms) to verify explicitly in every step of the isogeny chain.

We note that by not taking a full elimination approach as above, it might be possible to derive a whole system of equations (including auxiliary variables) that results in lower added complexity. The overhead compared to assuming Problem 2 is hard, will still be large though.

B Random walk graph

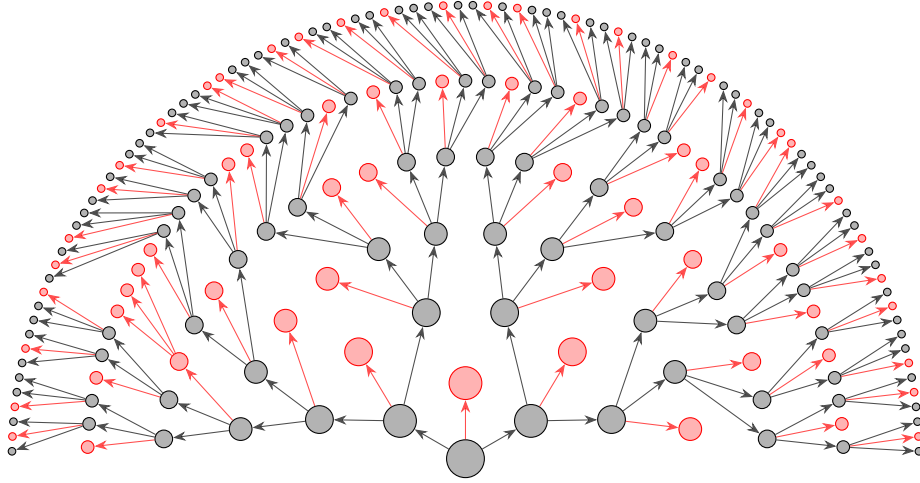


Fig. 1: Random walks starting from a superspecial theta null point of an abelian threefold over $\mathbb{F}_{p^2}$ for $p = 2^{61} - 1$. Red vertices correspond to tuples that do not correspond to abelian varieties. Note that from each vertex, we only choose 3 outgoing paths, if possible. This is for illustrative purposes (in total, there are $2^{2^g - 1} = 2^7$ valid outgoing paths, if there are any).

To test the heuristic hardness analysis of Problem 2 given in Section 4.6, we ran a depth-first search for solutions to Problem 2 in low characteristic for superspecial abelian threefolds. In practice, we did this by traversing the full graph of pseudo-paths starting from E_0^3 , for $E_0 : y^2 = x^3 + x$, and building the subgraph consisting of only isogeny-paths along the way (i.e. the edges and vertices in black in Figure 1). Table 5 reports the results, which strongly suggest that the complexity of such a brute-force search grows very quickly with the characteristic p .

	$p = 3$	$p = 7$	$p = 11$	$p = 19$
Avg. #vertices visited	328	9994	72442	852349
Median #vertices visited	230	1361	16554	514930

Table 5: Complexity of depth-first search against Problem 2 in dim. $g = 3$ over 100 runs.

Further, we also ran several breadth-first searches around singular theta null points (such as those with extra zeroes, or those coming from reducible varieties) under the hypothesis that such theta null points could be a source of solutions to Problem 2, but these searches did not yield any solutions in higher characteristic.

C Proof of Knowledge protocols

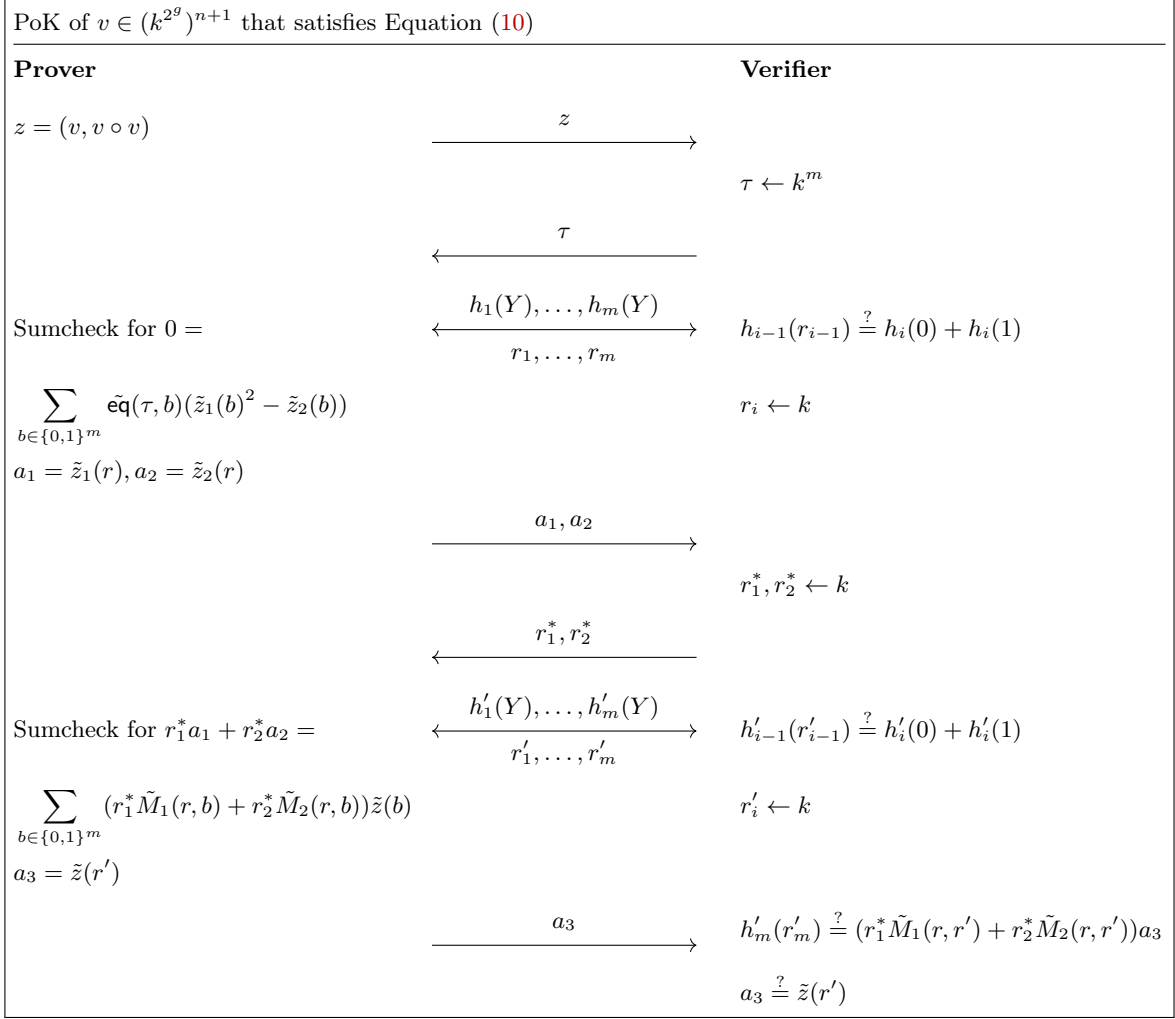


Fig. 2: Multilinear PIOP for Equation (11)

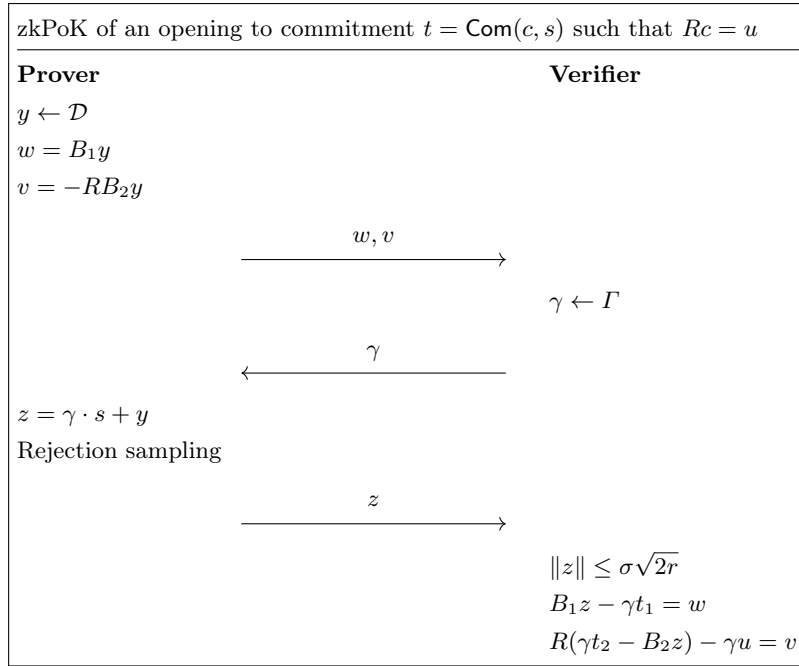


Fig. 3: Lattice-based zero-knowledge proof of knowledge with linear relation

zkPoK of $v \in (k^{2g})^{n+1}$ that satisfies Equation (10)

Prover		Verifier
$t_1 = B_1 s, t_{2,1} = B_{2,1} s + z$	$\xrightarrow{t_1, t_{2,1}}$	
	$\xleftarrow{\tau}$	$\tau \leftarrow k^m$
..... Sumcheck for $\sum_{b \in \{0,1\}^m} \tilde{\mathbf{eq}}(\tau, b)(\tilde{z}_1(b)^2 - \tilde{z}_2(b)) = 0$		
		$h_0(r_0) := 0$
$t_{2,i} = B_{2,i} s + (h_i(-1), h_i(0), h_i(1), h_i(2))$	$\xrightarrow{t_{2,i}}$	$r_i \leftarrow k$
	$\xleftarrow{r_i}$	Append $R, u : h_i(0) + h_i(1) \stackrel{?}{=} h_{i-1}(r_{i-1})$
.....		
$a_3 \leftarrow k, t_{2,m+1} = B_{2,m+1} s + (\tilde{z}_1(r), \tilde{z}_1(r)^2, a_3, \tilde{z}_2(r))$	$\xrightarrow{t_{2,m+1}}$	Append $R, u : a'_1 \stackrel{?}{=} a_2 + h_m(r_m)/\tilde{\mathbf{eq}}(\tau, r)$
$t_{2,m+2} = B_{2,m+2} s + (q_1, \dots, q_9)$	$\xrightarrow{t_{2,m+2}}$	
	$\xleftarrow{y}$	$y \leftarrow k$
	$\xrightarrow{p(y), q(y)}$	$q(y) \stackrel{?}{=} p(y)(p(y) - 2y^{-2})$
		Append $R, u : p(y) \stackrel{?}{=} p _y, q(y) \stackrel{?}{=} q _y$
	$\xleftarrow{r_1^*, r_2^*}$	$r_1^*, r_2^* \leftarrow k$
..... Sumcheck for $r_1^* \tilde{z}_1(r) + r_2^* \tilde{z}_2(r) = \sum_{b \in \{0,1\}^m} (r_1^* \tilde{M}_1(r, b) + r_2^* \tilde{M}_2(r, b)) \tilde{z}(b)$		
		$h'_0(r_0) := r_1^* \tilde{z}_1(r) + r_2^* \tilde{z}_2(r)$
$t_{2,m+2+i} = B_{2,m+2+i} s + (h'_i(-1), h'_i(0), h'_i(1))$	$\xrightarrow{t_{2,m+2+i}}$	$r'_i \leftarrow k$
	$\xleftarrow{r'_i}$	Append $R, u : h'_i(0) + h'_i(1) \stackrel{?}{=} h'_{i-1}(r'_{i-1})$
.....		
$t_{2,2m+3} = B_{2,2m+3} s + \tilde{z}(r')$	$\xrightarrow{t_{2,2m+3}}$	Append $R, u : \tilde{z}(r') \stackrel{?}{=} \tilde{z} _{r'}$
$y \leftarrow \mathcal{D}, w = B_1 y, v = -RB_2 y$	$\xrightarrow{w, v}$	
	$\xleftarrow{\gamma}$	$\gamma \leftarrow \Gamma$
$z = \gamma \cdot s + y$, Rejection sampling	$\xrightarrow{z}$	$\ z\ \leq \sigma\sqrt{2r}$
		$B_1 z - \gamma t_1 = w$
		$R(\gamma t_2 - B_2 z) - \gamma u = v$

Fig. 4: Zero-knowledge proof system for Equation (10) with small proof size